GE Healthcare

LightSpeed 5.X

Functional Interconnects

5113145-100
Rev 04

This Page Intentionally Blank.

IMPORTANT PRECAUTIONS

LANGUAGE

WARNING

- **THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.**
- **IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.**
- **DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.**
- **FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.**

AVERTISSEMENT

- **CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.**
- **SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.**
- **NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.**
- **LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.**

WARNUNG

- **DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.**
- **FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.**
- **VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.**
- **WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.**

AVISO

- **ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.**
- **SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.**
- **NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.**
- **LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.**

ATENÇÃO

- **ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.**
- **SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.**
- **NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.**
- **O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A' CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.**

AVVERTENZA

- **IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.**
- **SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.**
- **SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.**
- **NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.**

警告

このサービスマニュアルには英語版しかありません。

ＧＥＭＳ以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。

このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。

この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意:

本维修手册仅存有英文本。

非 GEMS 公司的维修员要求非英文本的维修手册时，

客户需自行负责翻译。

未详细阅读和完全了解本手册之前，不得进行维修。

忽略本注意事项会对维修员，操作员或病人造成触

电，机械伤害或其他伤害。

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "damage in shipment" written on all copies of the freight or express bill before delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

To file a report:

- Call 1-800-548-3366 and use option 8.
- Fill out a report on http://us44hdd21/sctq/InstallFulfill/InstalFulfillment.htm
- Contact your local service coordinator for more information on this process.

Rev. Jan. 5, 2005

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Healthcare personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

LITHIUM BATTERY CAUTIONARY STATEMENTS

CAUTION ⚠

Risk of Explosion

Danger of explosion if battery is incorrectly replaced.

Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer's instructions.

ATTENTION ⚠

Danger d'Explosion

Il y a danger d'explosion s'il y a replacement incorrect de la batterie.

Remplacer uniquement avec une batterie du même type ou d'un type recommandé par le constructeur. Mettre au rébut les batteries usagées conformément aux instructions du fabricant.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the GE Healthcare PQR process to report all omissions, errors, and defects in this publication.

Revision History

Rev	Date	Reason for change
1	06/07/04	Initial Release
2	04/21/05	PQR 13034160; PQR 13034162
3	07/21/05	PQR 13036875
4	07/18/06	PQR 13031439

Table of Contents

Chapter 1
System Communications

Section 1.0 — Slip Ring Communications Interface - 2399621

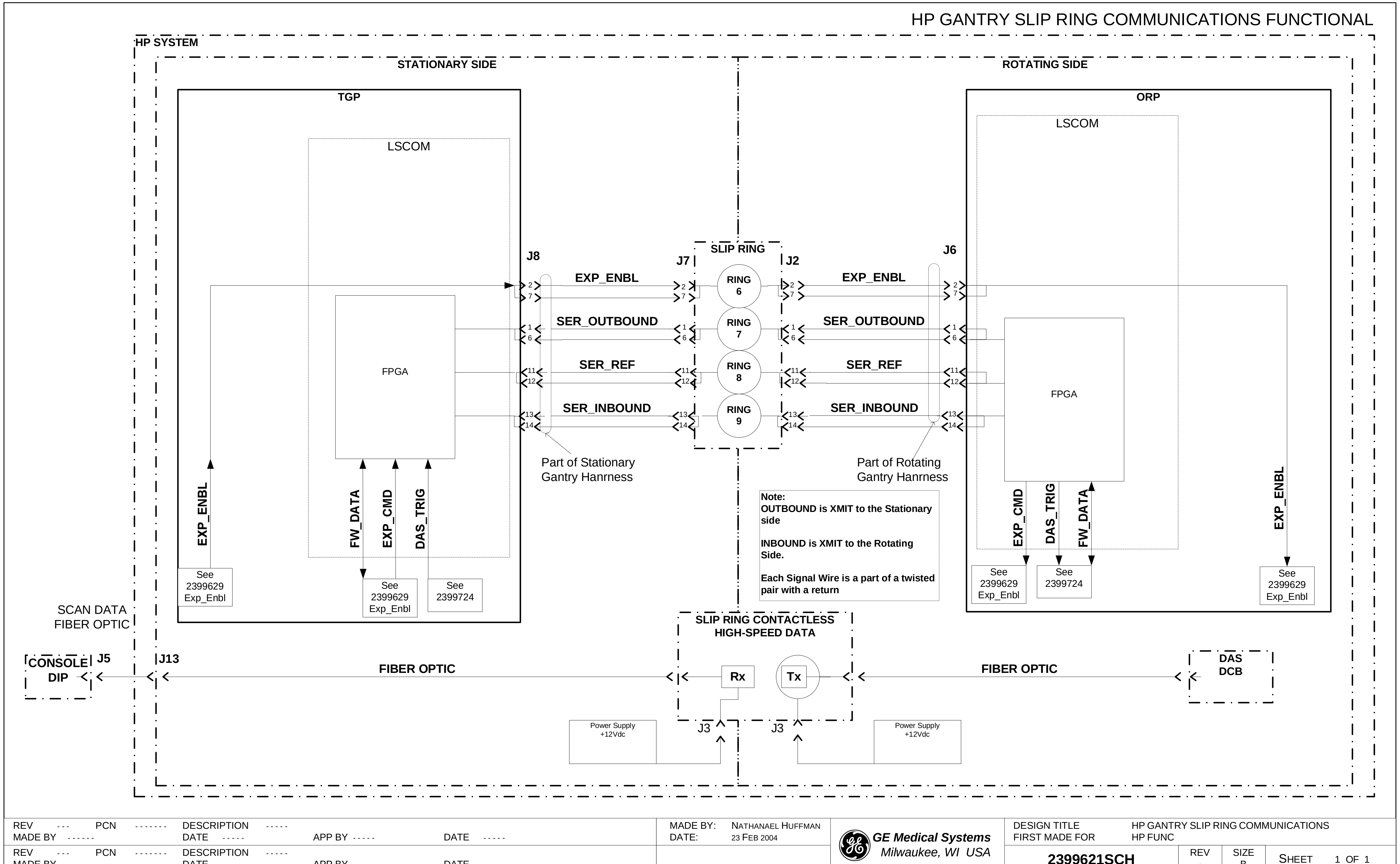


Figure 1-1 2399621 Sheet 1, Rev. 1

Section 2.0 — Final Scan Control, Console Ethernet - 2399622

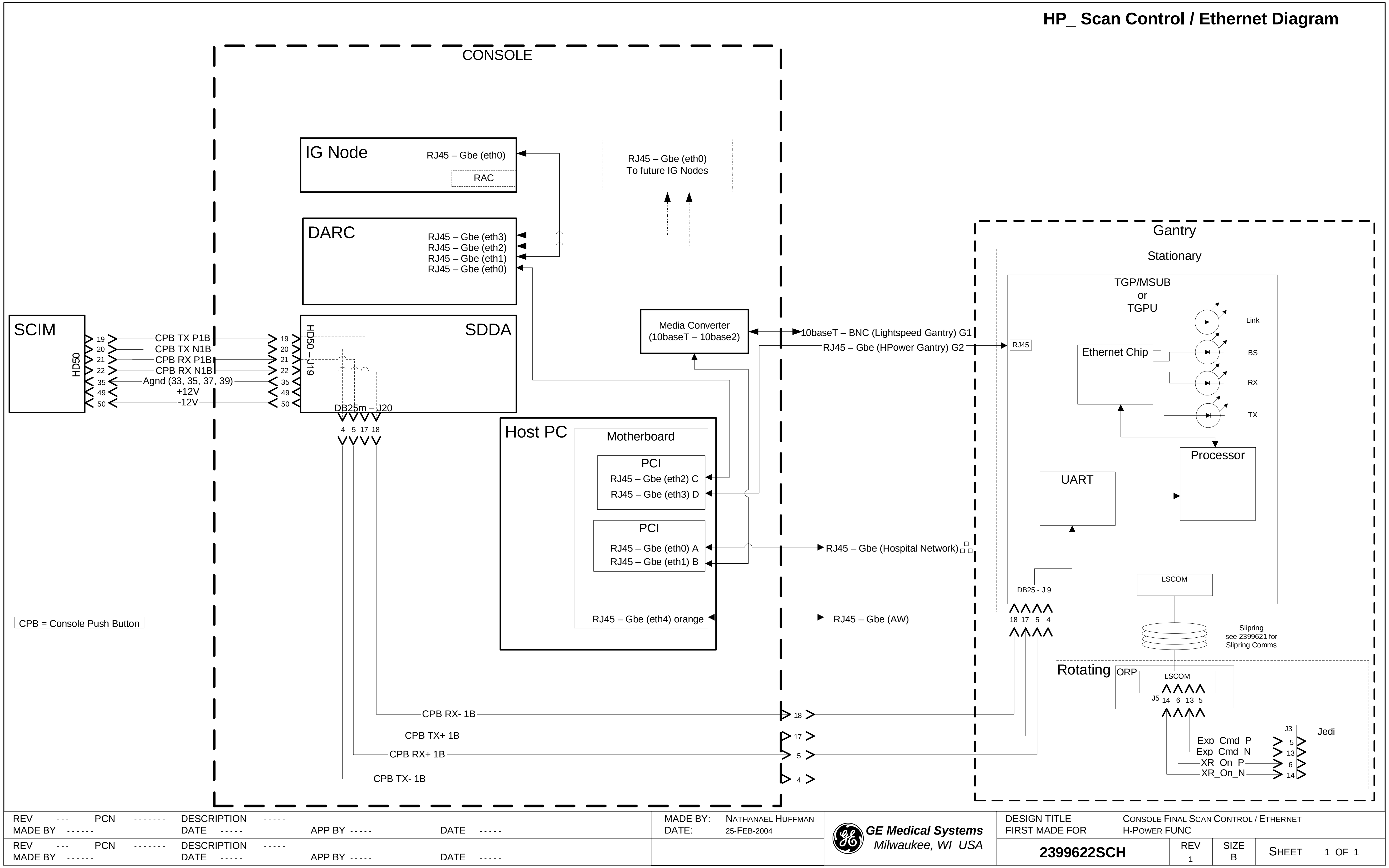
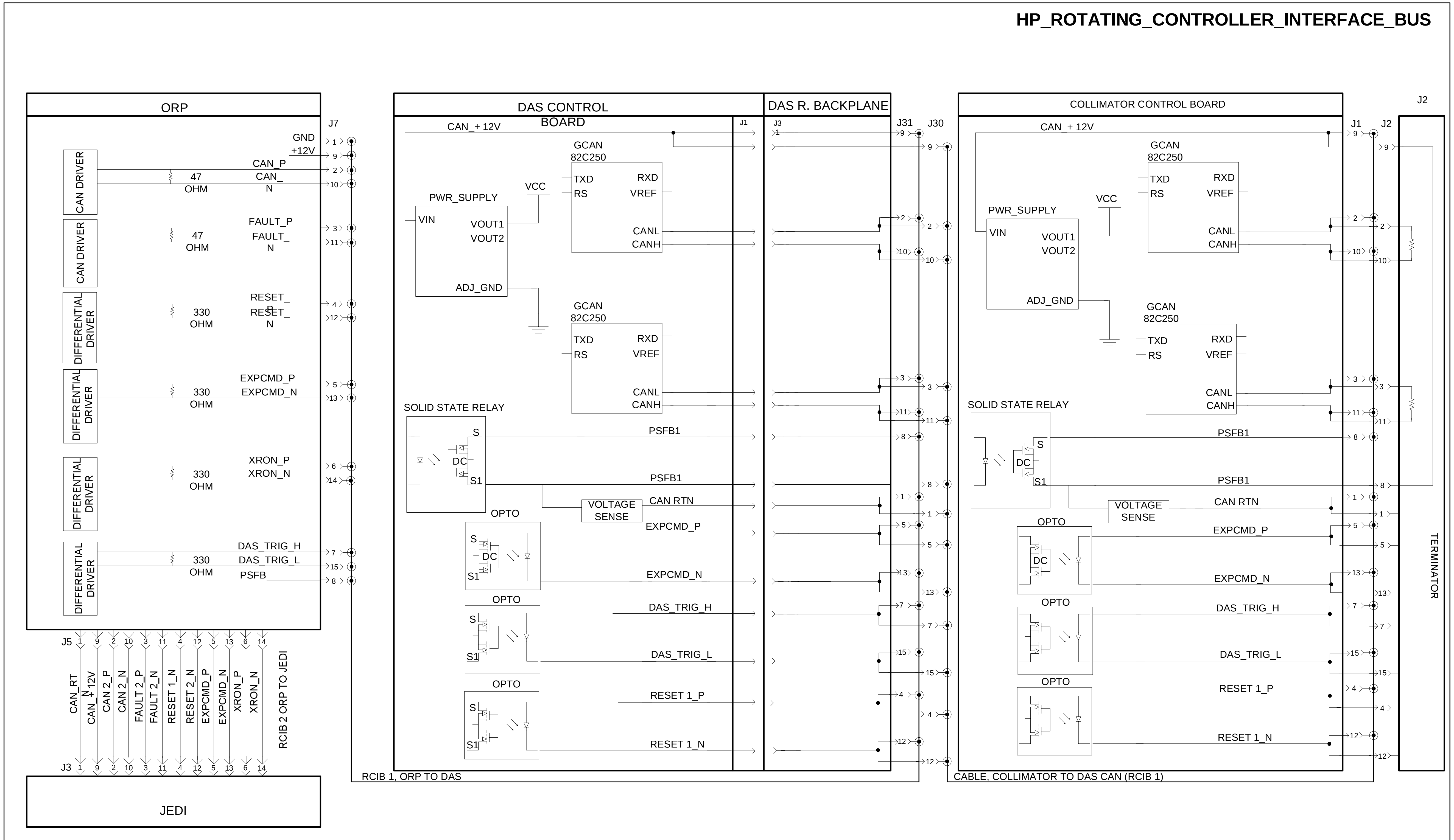


Figure 1-2 2399622 Sheet 1, Rev 1

Section 3.0 — Rotating Controller Interface Bus (RCIB) - 2399623



REV ---	PCN -------	DESCRIPTION -----			MADE BY: ANDY PANOS	GE Medical Systems Milwaukee, WI USA	DESIGN TITLE HP_ROTATING_CONTROLLER_INTERFACE_BUS	
MADE BY ------		DATE -----	APP BY -----	DATE -----	DATE: 03/05/04		FIRST MADE FOR H-POWER FUNC	
REV ---	PCN -------	DESCRIPTION -----					2399623SCH	REV 1 / SIZE B / SHEET 1 OF 1
MADE BY ------		DATE -----	APP BY -----	DATE -----				

Figure 1-3 2399623 Sheet 1, Rev. 1

Chapter 2
X-Ray Generation

Section 1.0 — Tube Fans / Pump / SmartID Interface - 2399624

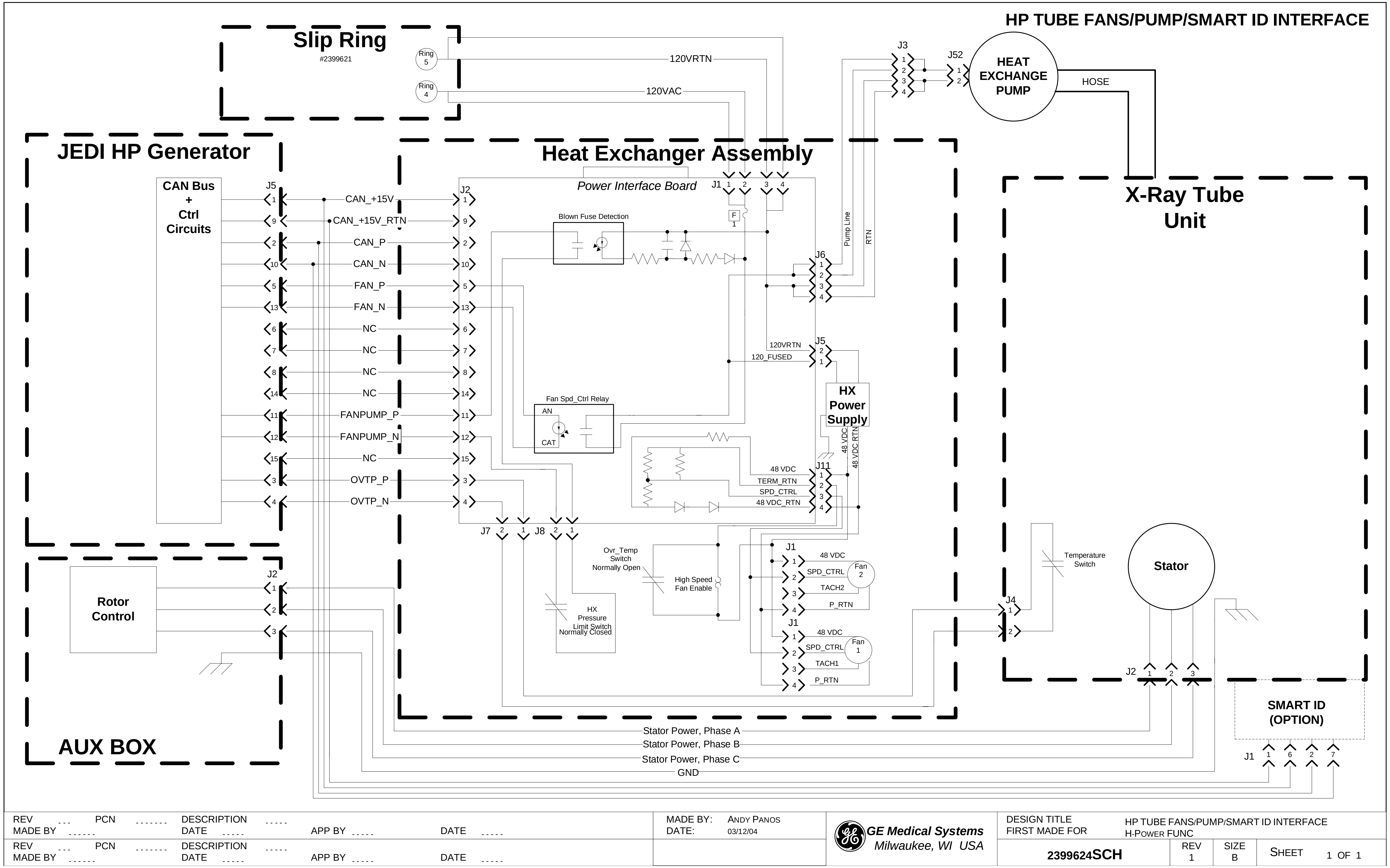
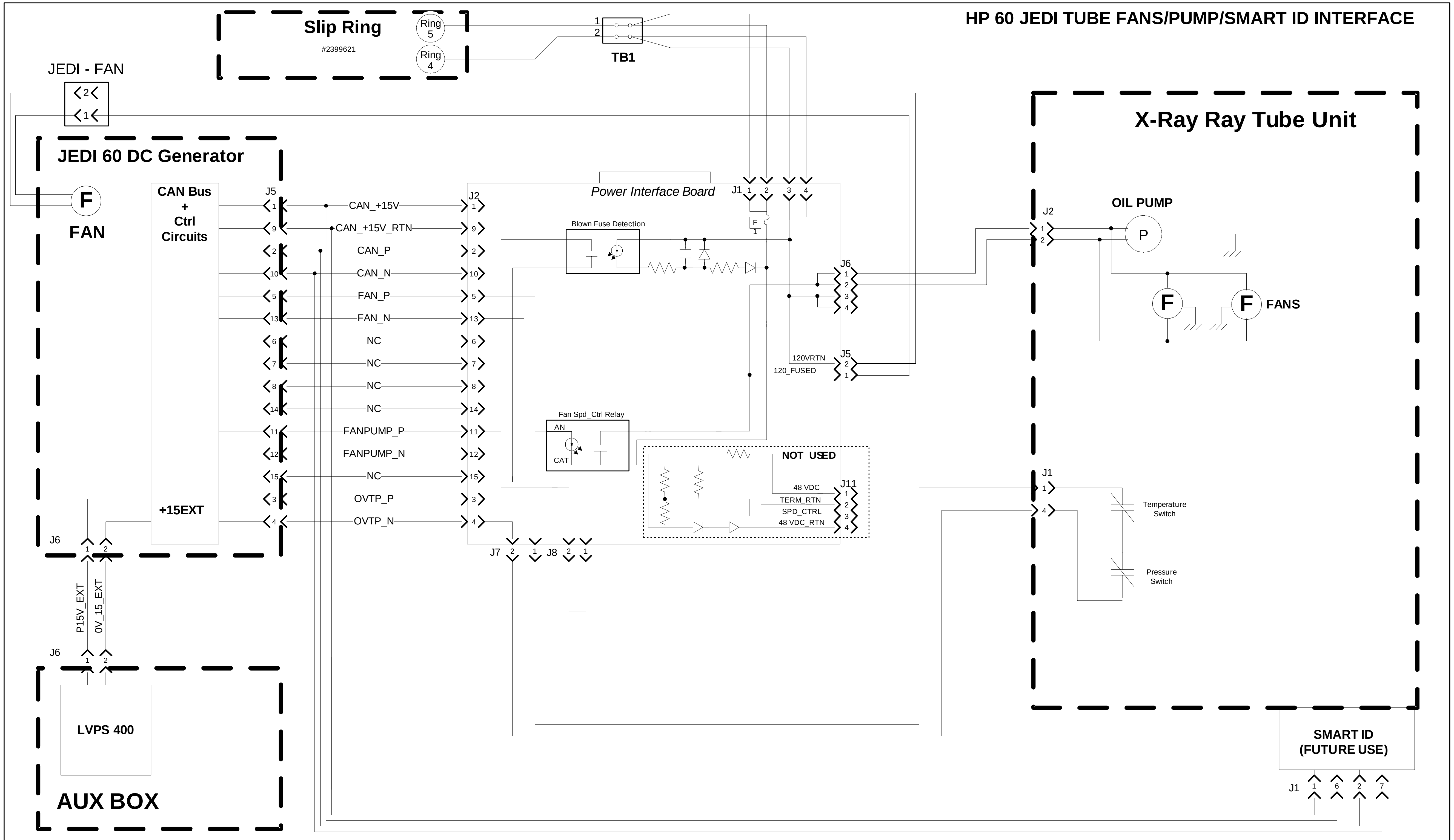


Figure 2-1 2399624 Sheet 1, Rev. 1

Section 2.0 — HP60 JEDI Tube Fans/Pump/Smart ID Interface - 5141517



REV --- PCN ------- MADE BY ------	DESCRIPTION ----- DATE ----- APP BY---- DATE -----	MADE BY: KENICHI ODAGIRI DATE: MAY/10/2005	GE Medical Systems Milwaukee, WI USA	DESIGN TITLE HP 60 JEDI TUBE FANS/PUMP/SMART ID INTERFACE FIRST MADE FOR H-POWER 60 FUNC	
REV --- PCN ------- MADE BY ------	DESCRIPTION ----- DATE ----- APP BY---- DATE -----	MADE BY: NAOKI ARIYAMAI DATE: MAY/10/2005		5141517SCH REV 1 SIZE B	SHEET 1 OF 1

Figure 2-2 5141517 Sheet 1, Rev 1

Section 3.0 — JEDI Interface - kV Loop - 2399625

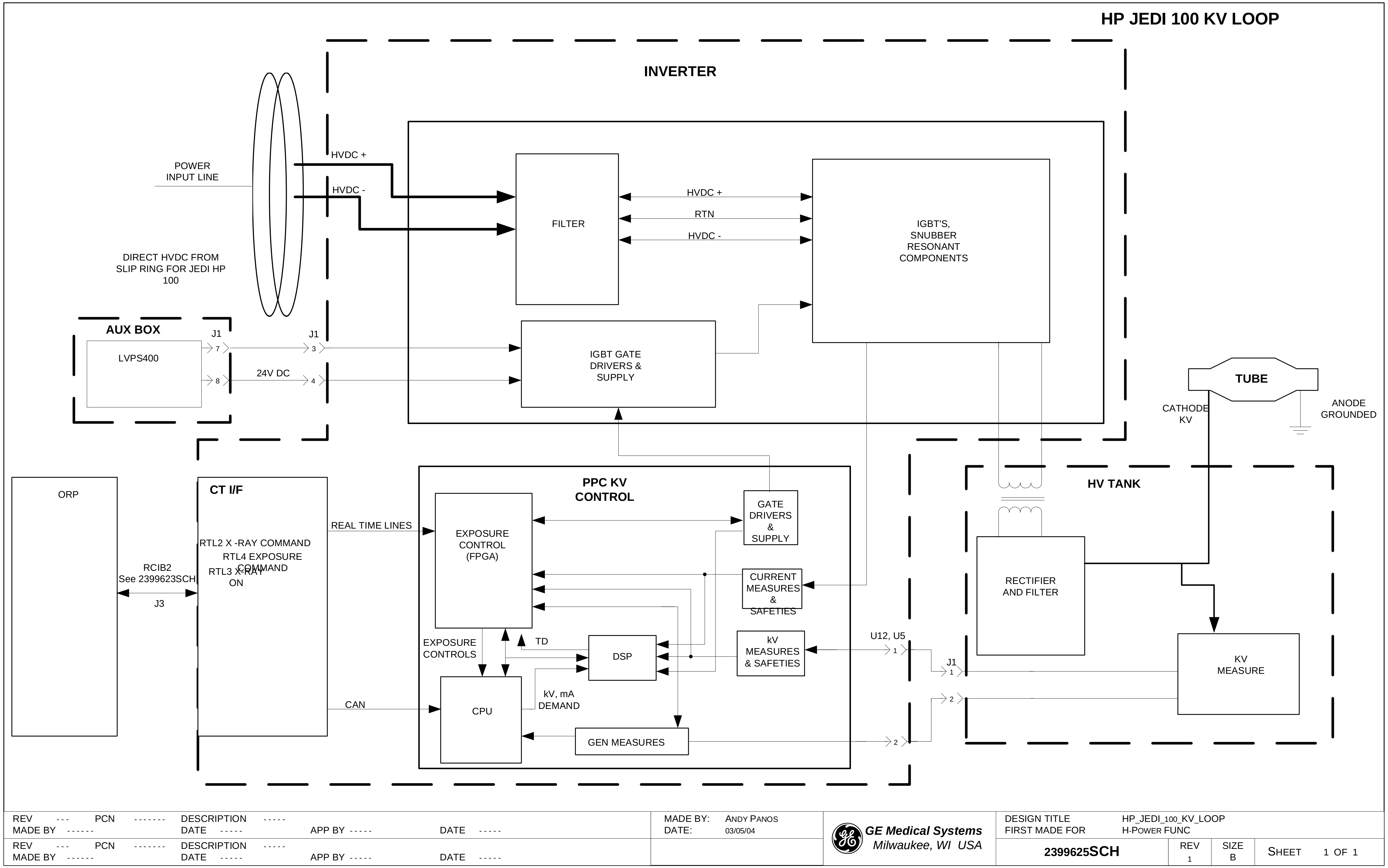
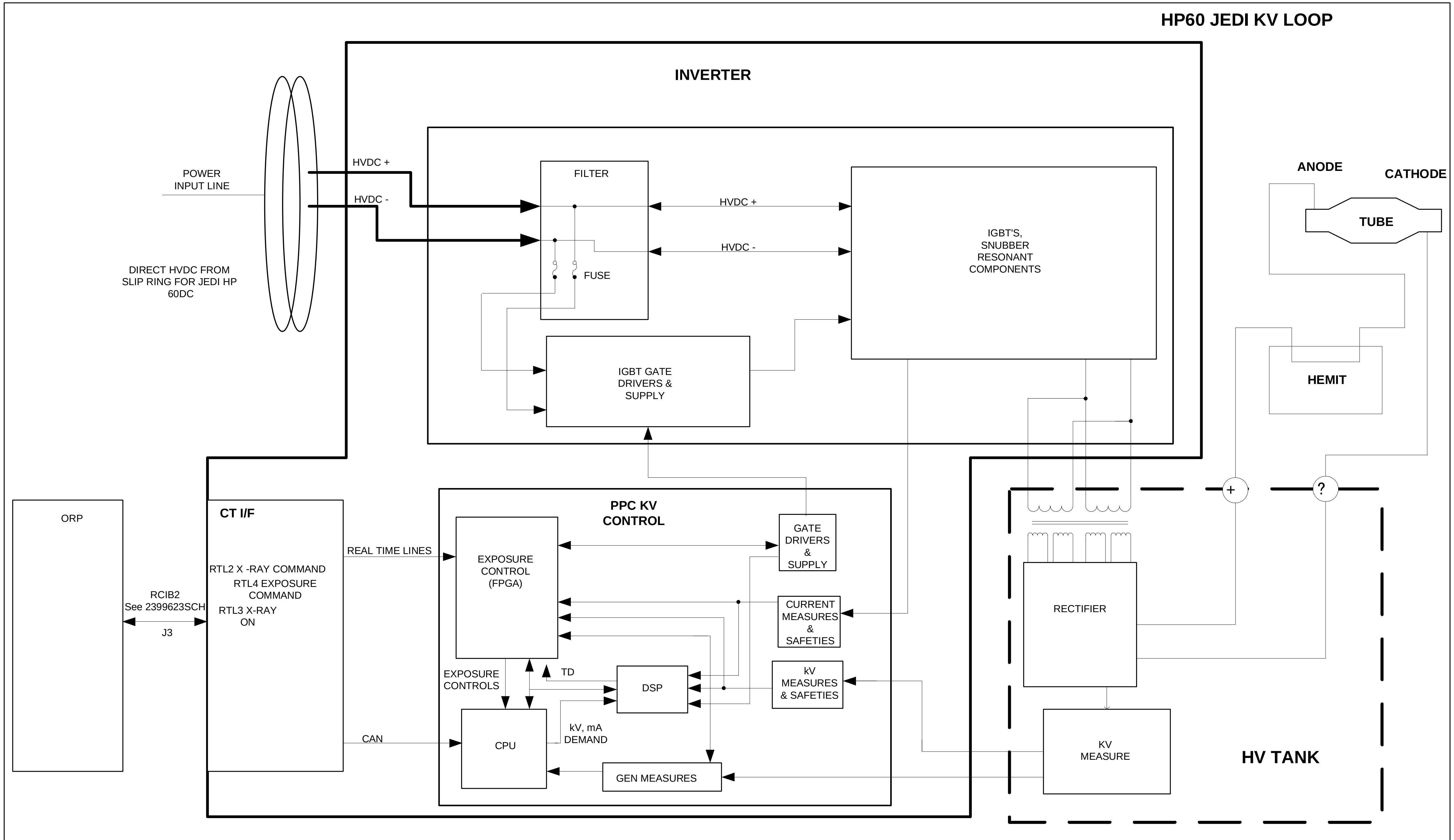


Figure 2-3 2399625 Sheet 1, Rev. 1

Section 4.0 — HP60 JEDI kV Loop - 5141524



REV - - -	PCN - - - - - - -	DESCRIPTION - - - - -			MADE BY: KENICHI ODAGIRI	GE Medical Systems Milwaukee, WI USA	DESIGN TITLE	HP60_JEDI_KV_LOOP	
MADE BY - - - - - -		DATE - - - - -	APP BY - - - - -	DATE - - - - -	DATE: MAY/10/05		FIRST MADE FOR	H-POWER FUNC	
REV - - -	PCN - - - - - - -	DESCRIPTION - - - - -			CHECK BY: NAOKI ARIYAMA		5141524SCH	REV 1	SIZE B
MADE BY - - - - - -		DATE - - - - -	APP BY - - - - -	DATE - - - - -	DATE: MAY/10/05				SHEET 1 OF 1

Figure 2-4 5141524 Sheet 1, Rev 1

Section 5.0 — JEDI Interface - mA Loop - 2399628

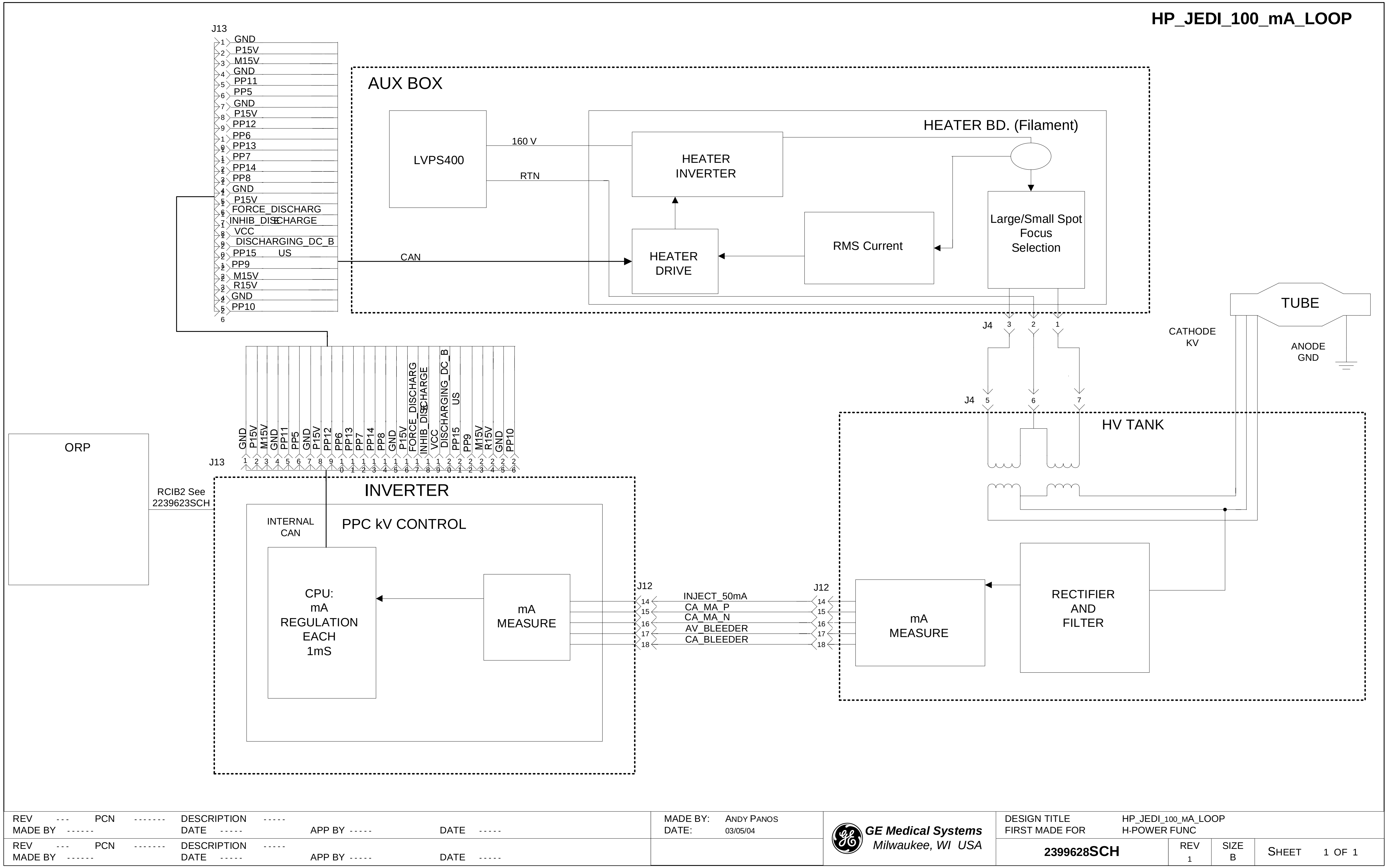


Figure 2-5 2399628 Sheet 1, Rev. 1

Section 6.0 — HP60 JEDI mA Loop - 5141527

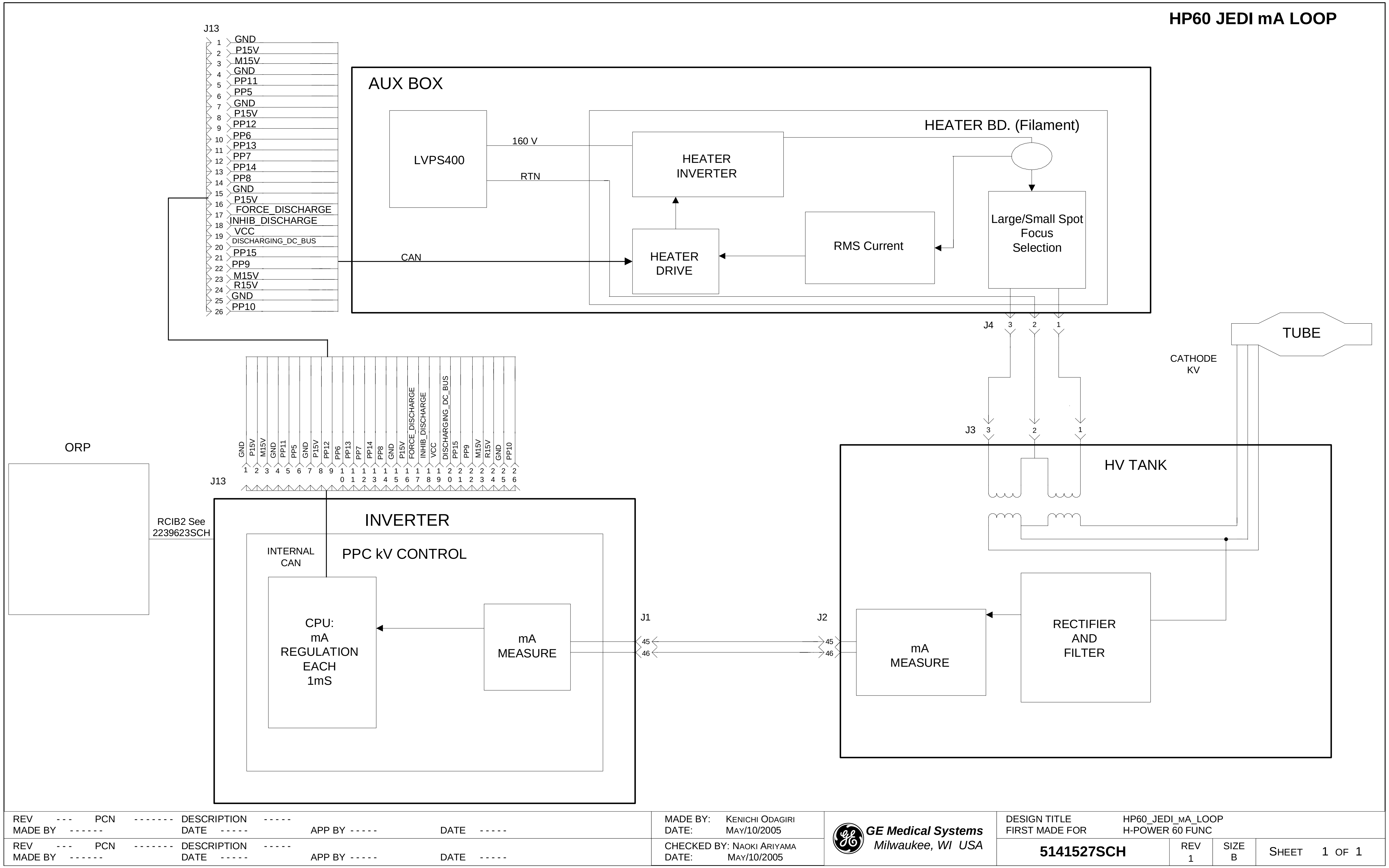


Figure 2-6 5141527 Sheet 1, Rev 1

Section 7.0 — X-Ray Collimation & Filtration - 2399626

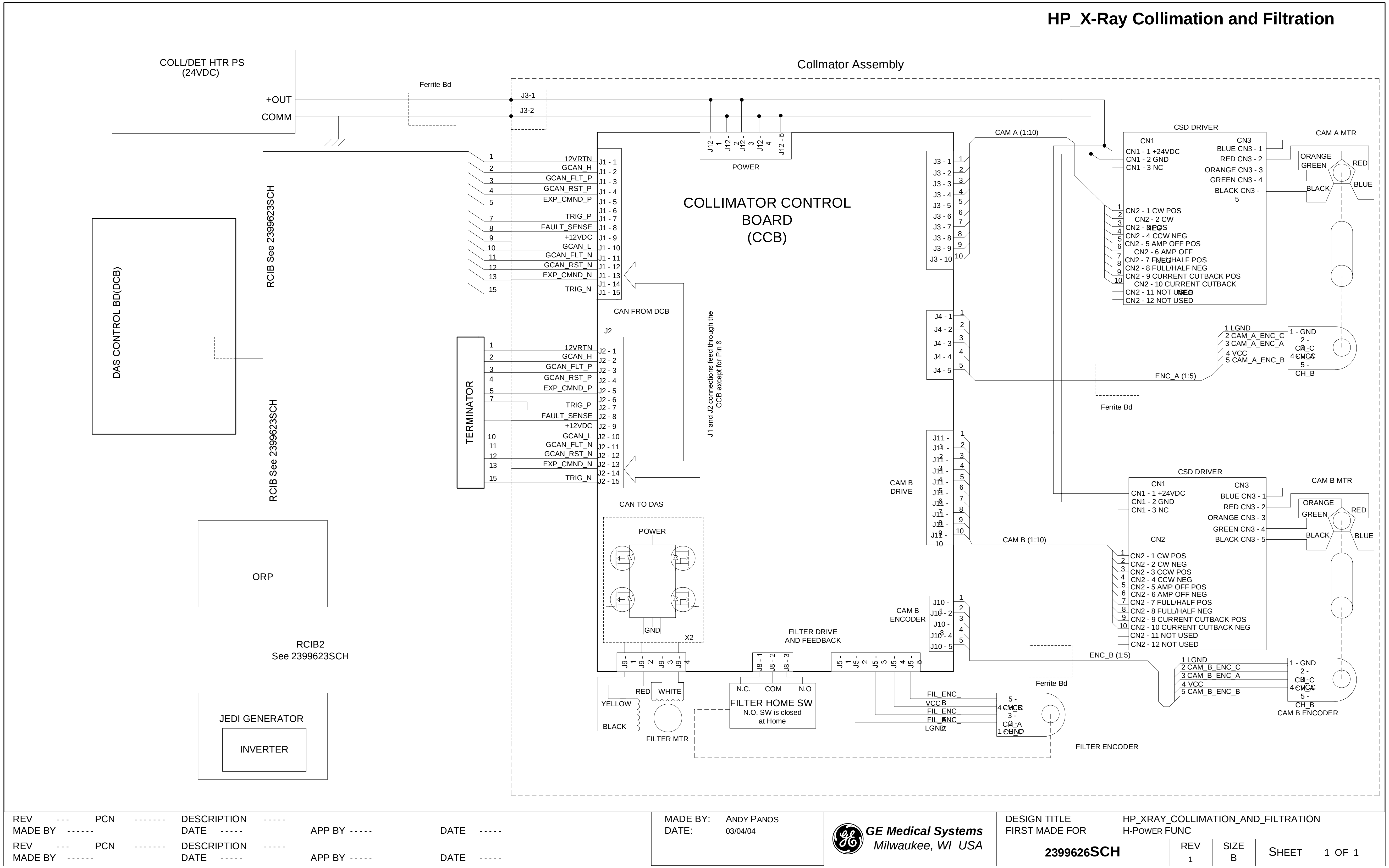


Figure 2-7 2399626 Sheet 1, Rev. 1

Section 8.0 — Exposure Enable and Final Command - 2399629

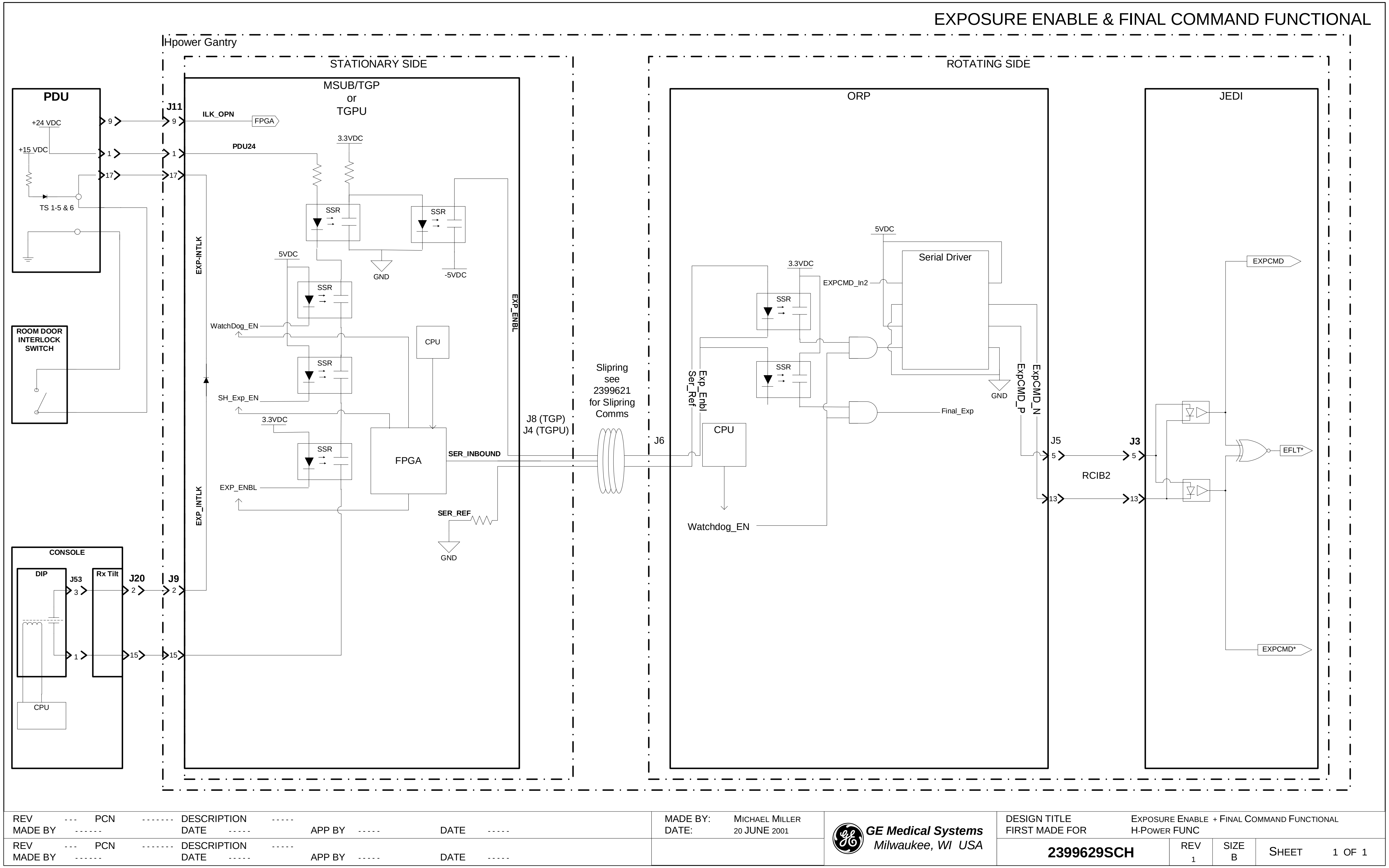


Figure 2-8 2399629 Sheet 1, Rev. 1

Section 9.0 — Tube Rotor Control - 2399630

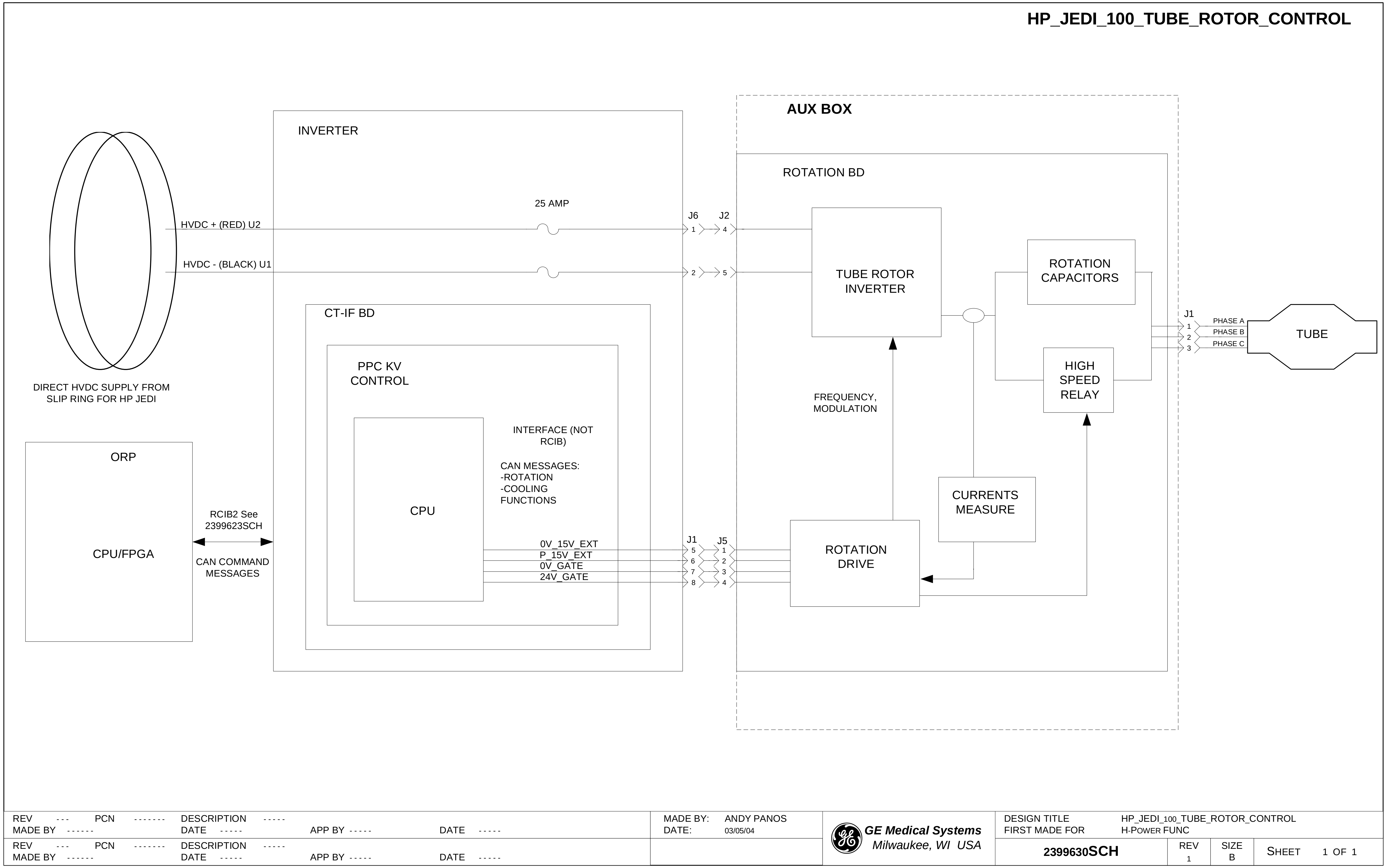


Figure 2-9 2399630 Sheet 1, Rev. 1

Section 10.0 — HP60 JEDI Tube Rotor Control - 5141528

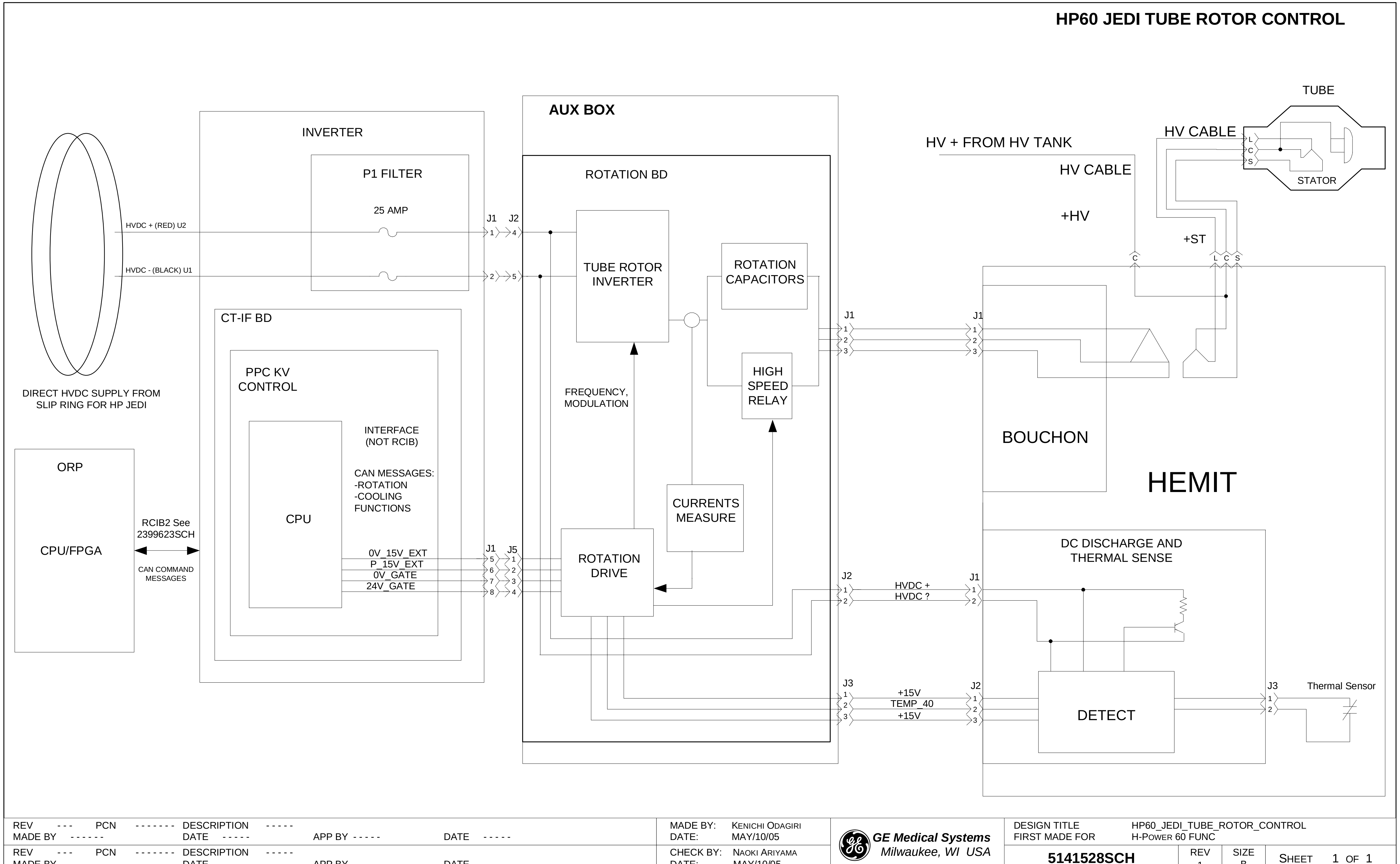


Figure 2-10 5141528 Sheet 1, Rev 1

Section 11.0 — HP X-Ray Tube Coolant Flow - 2399628

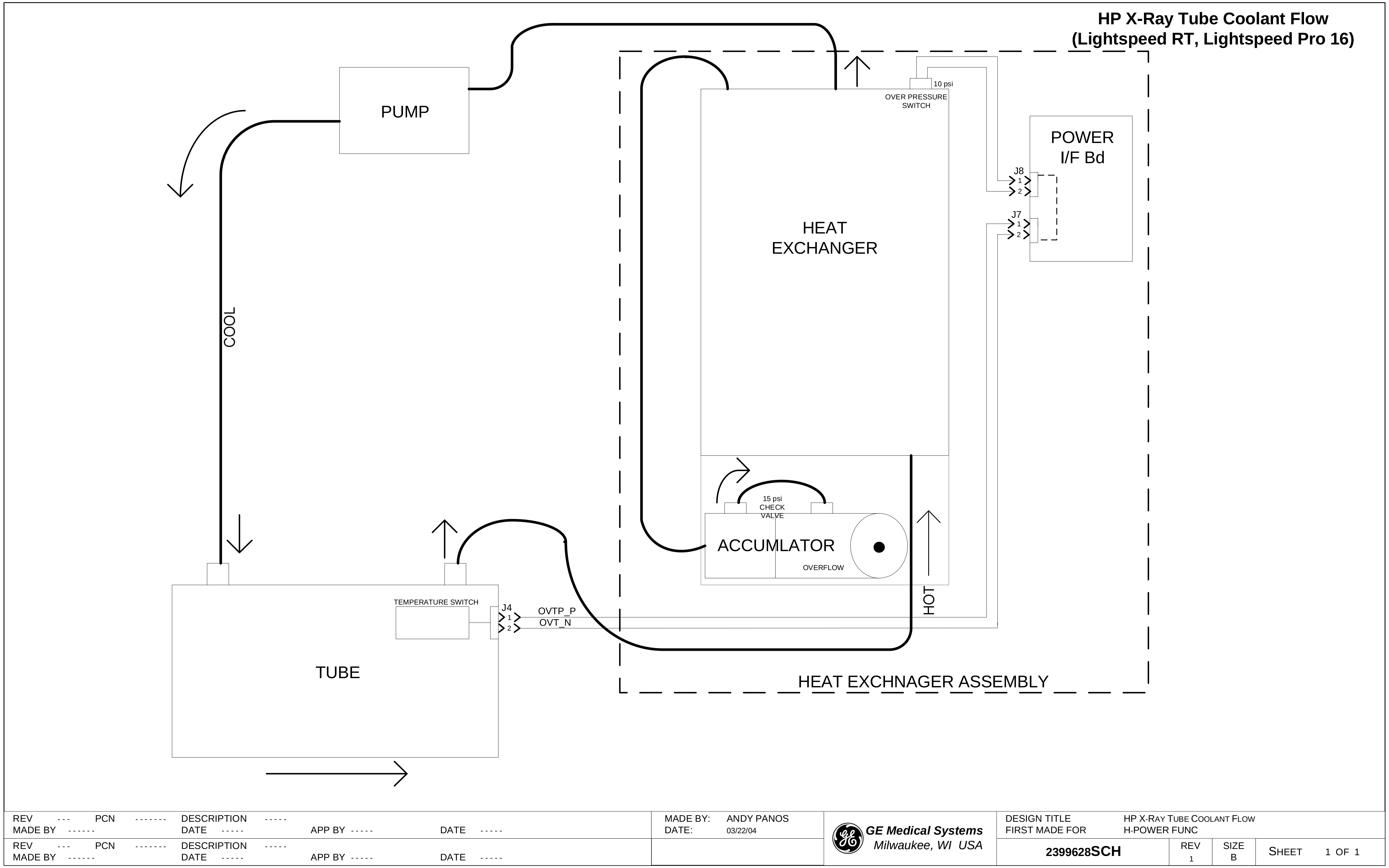


Figure 2-11 2399628 Sheet 1, Rev. 1

Chapter 3
Data Acquisition

Section 1.0 — Trigger Generation - 2399724

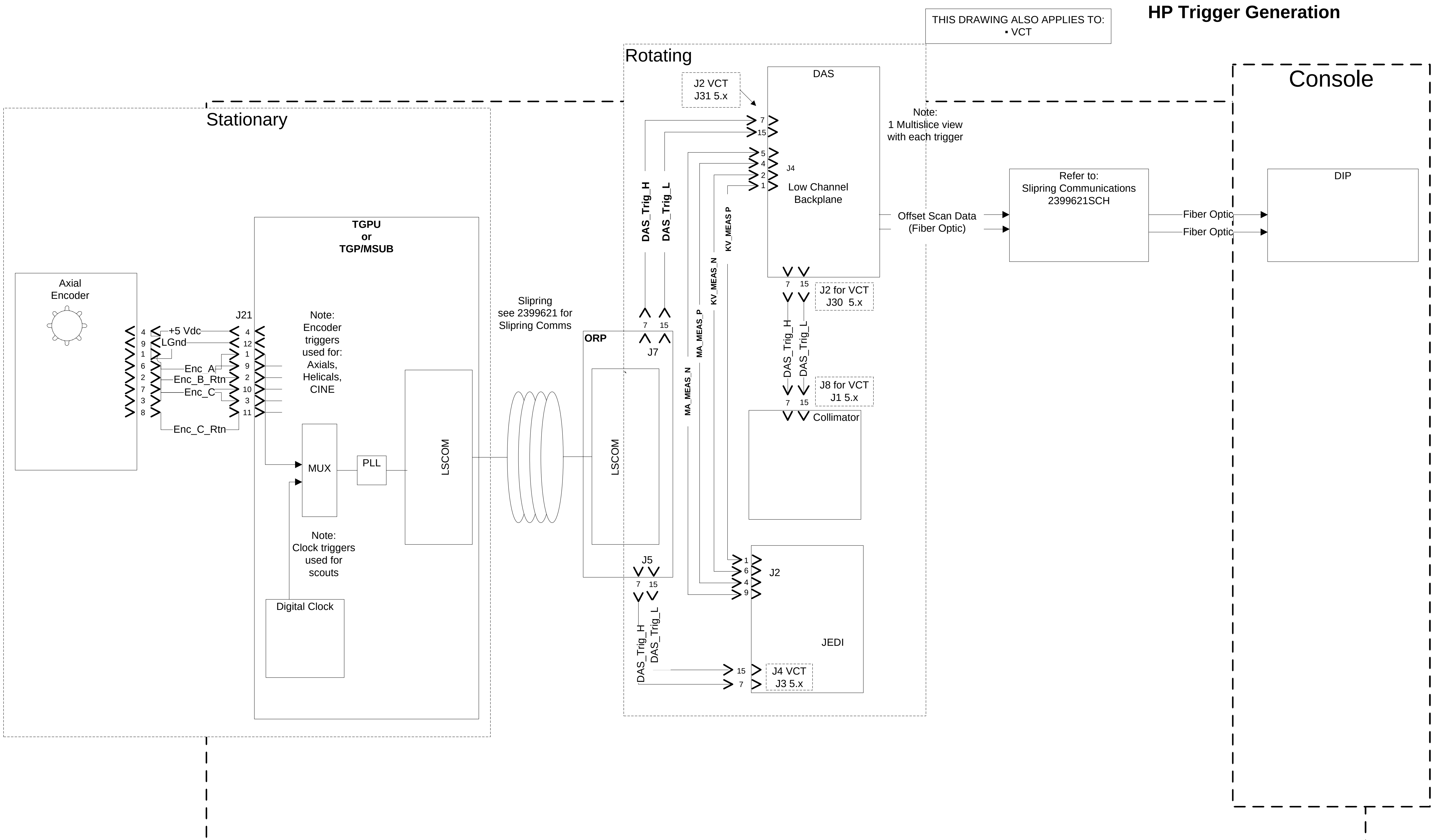


Figure 3-1 2399724 Sheet 1

Section 2.0 — Detector Heater (DAS Item) - 2399725

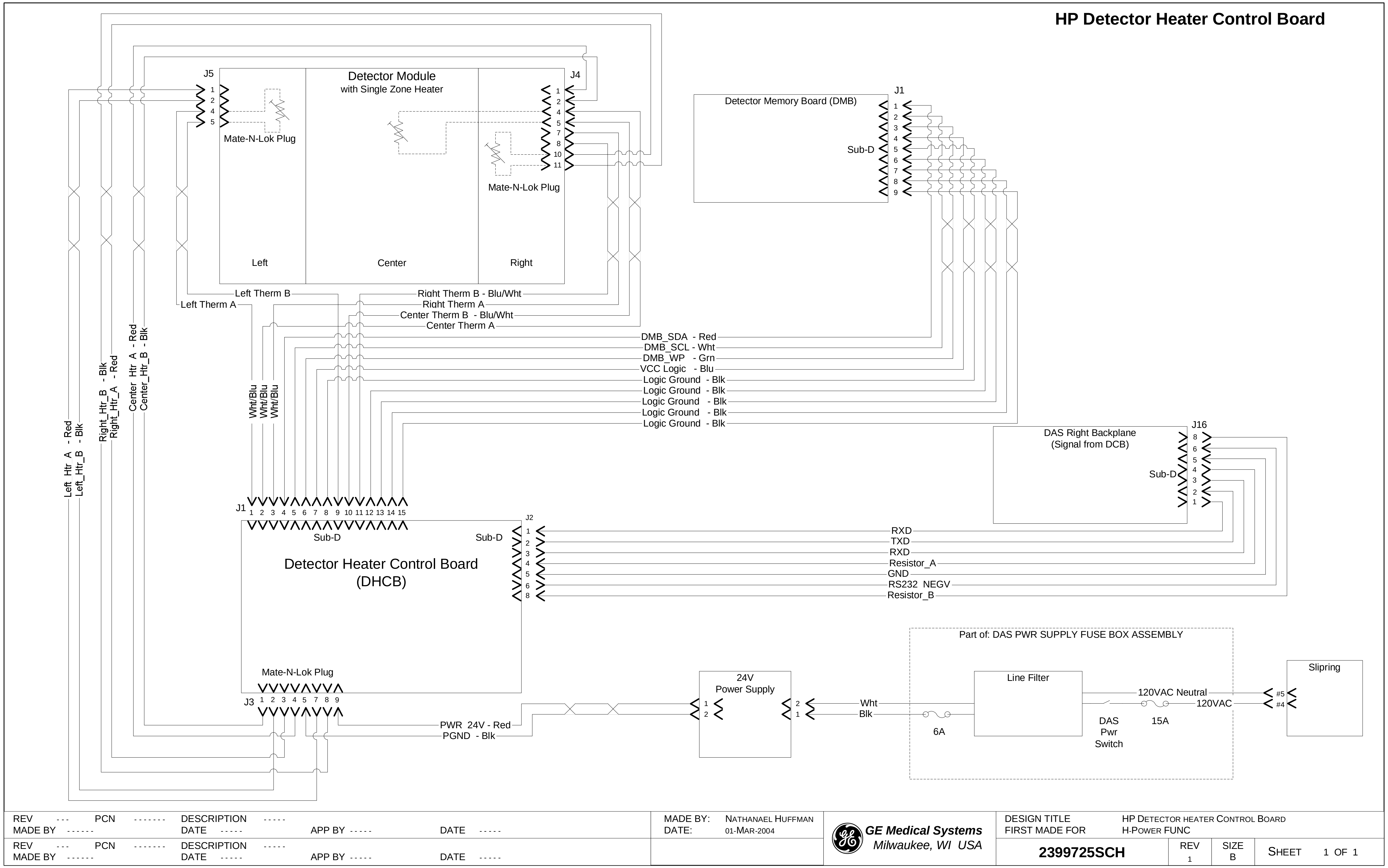
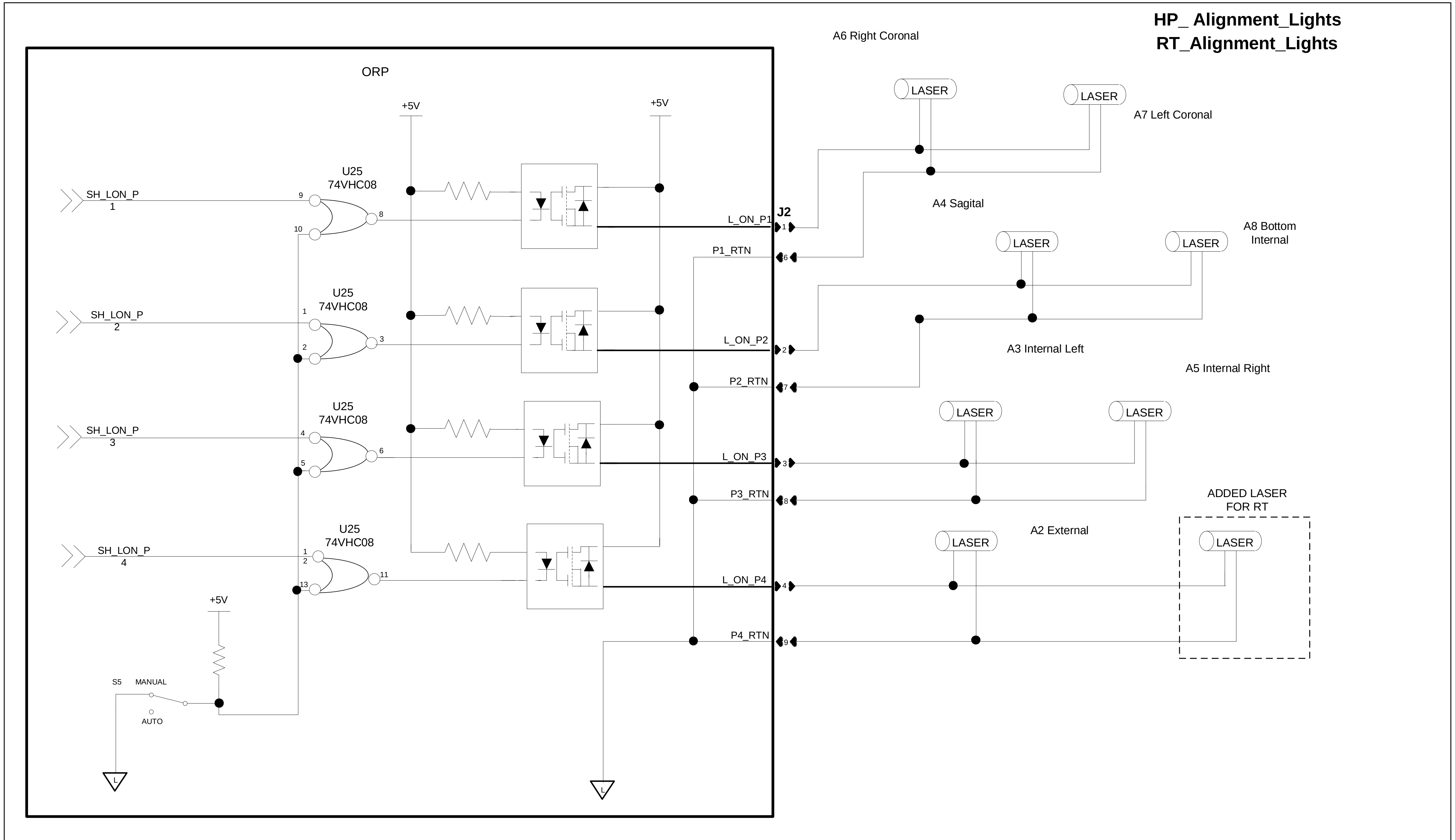


Figure 3-2 2399725 Sheet 1, Rev. 1

This Page Intentionally Blank.

Chapter 4
Patient Positioning

Section 1.0 — Alignment Lights - 2399726



REV --- PCN ------- DESCRIPTION ----- MADE BY ------ DATE ----- APP BY ----- DATE -----	MADE BY: ANDY PANOS DATE: 03/17/04	GE Medical Systems Milwaukee, WI USA	DESIGN TITLE HP_ALIGNTMENT_LIGHTS FIRST MADE FOR H-POWER FUNC		
REV --- PCN ------- DESCRIPTION ----- MADE BY ------ DATE ----- APP BY ----- DATE -----			2399726SCH	REV 1	SIZE B / SHEET 1 OF 1

Figure 4-1 2399726 Sheet 1, Rev 1

Section 2.0 — Gantry Tilt, from Gantry, Console - 2399727

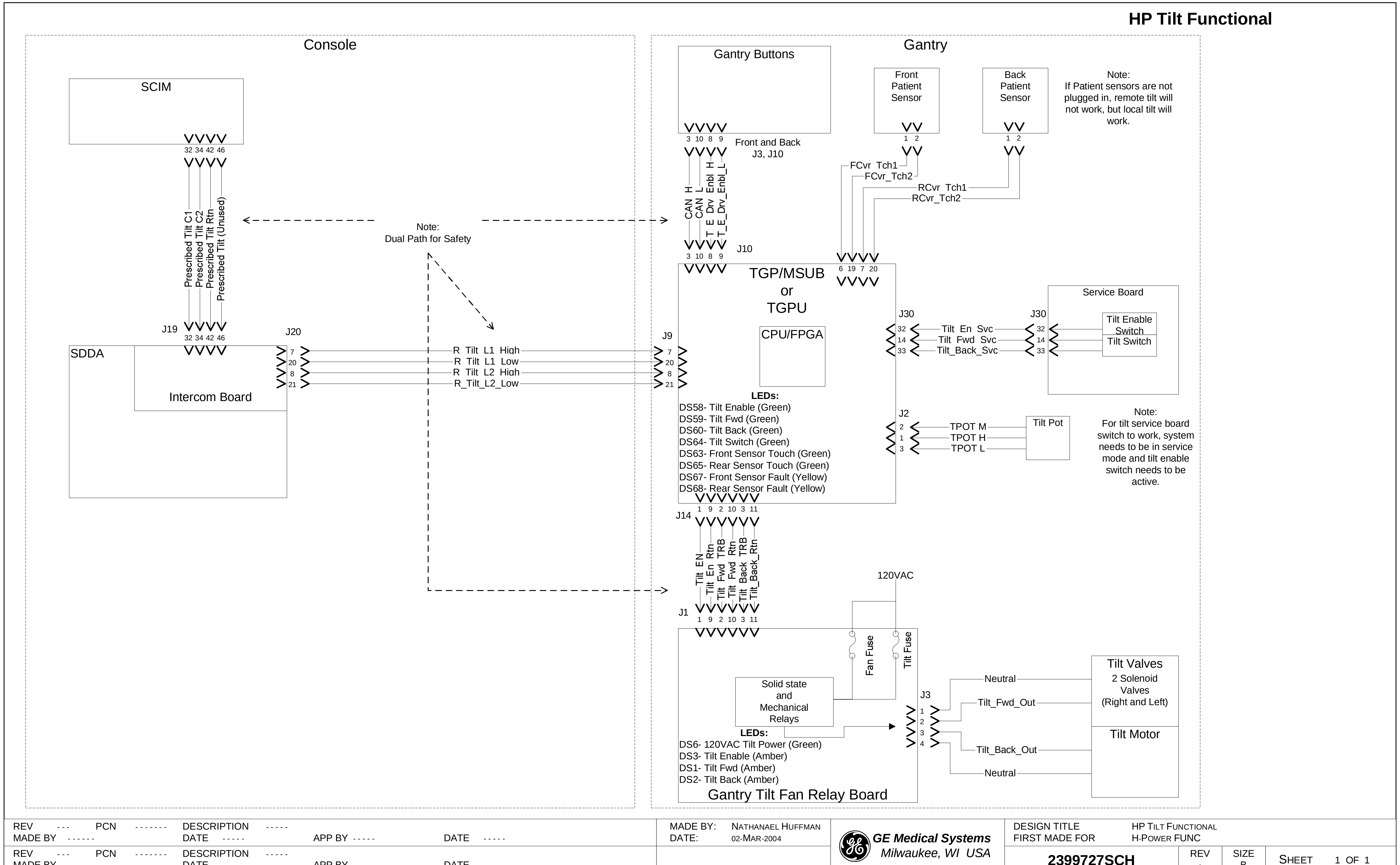
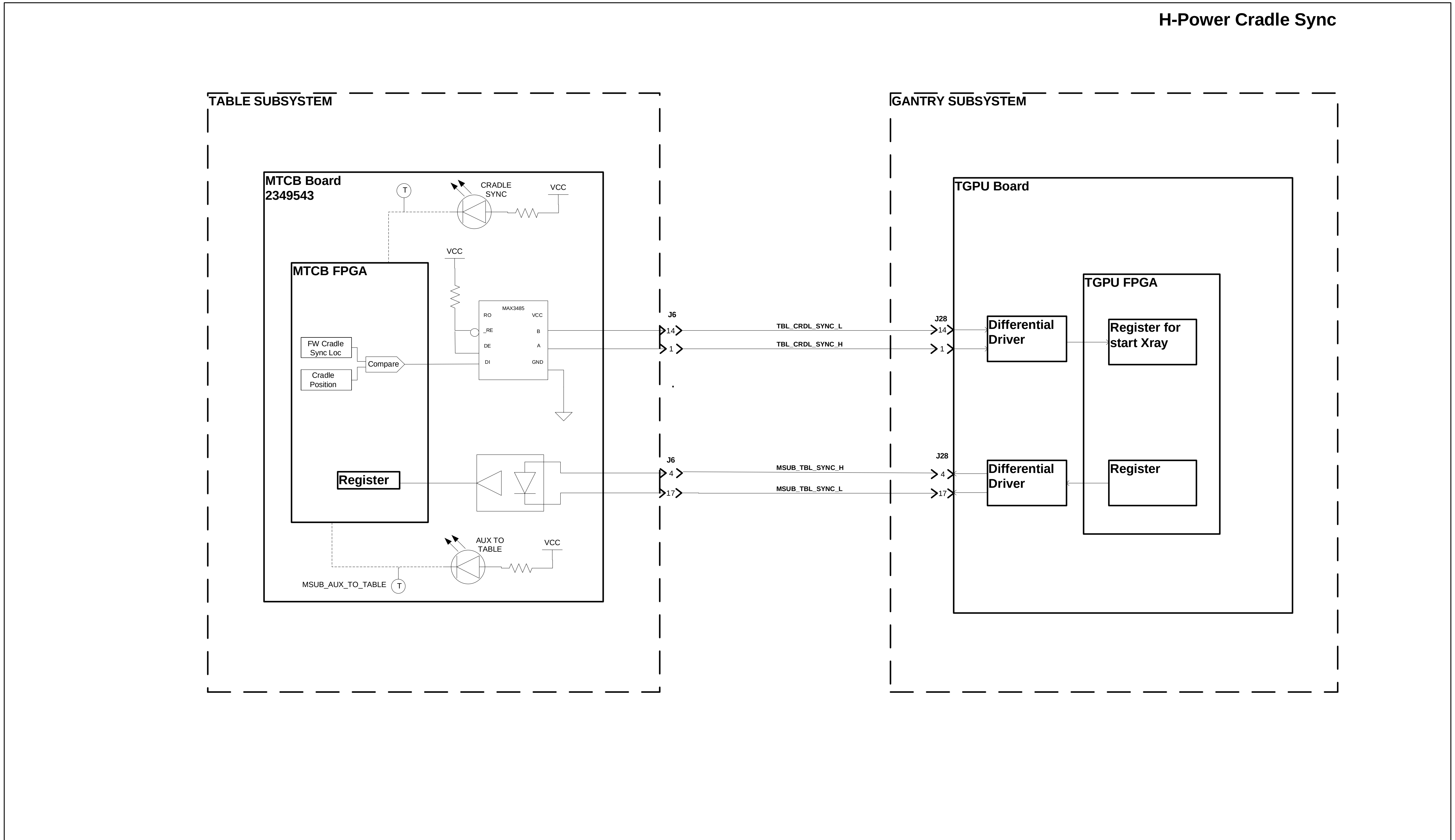


Figure 4-2 2399727 Sheet 1, Rev. 1

Section 3.0 — Table Cradle Sync - 2399728



REV --- PCN ------- DESCRIPTION ----- MADE BY ------ DATE ----- APP BY ----- DATE -----	MADE BY: GEORGE CAO DATE: NOV 18, 2003	GE Medical Systems Milwaukee, WI USA	DESIGN TITLE H-POWER CRADLE SYNC FIRST MADE FOR H-POWER FUNC		
REV --- PCN ------- DESCRIPTION ----- MADE BY ------ DATE ----- APP BY ----- DATE -----	ELECTRONIC RELEASE PROCESS **NOT** DESIGN RELEASED		2399728SCH	REV 1	SIZE B	SHEET 1 OF 1

Figure 4-3 2399728 Sheet 1, Rev. 1

Section 4.0 — Table Elevation - 2399729

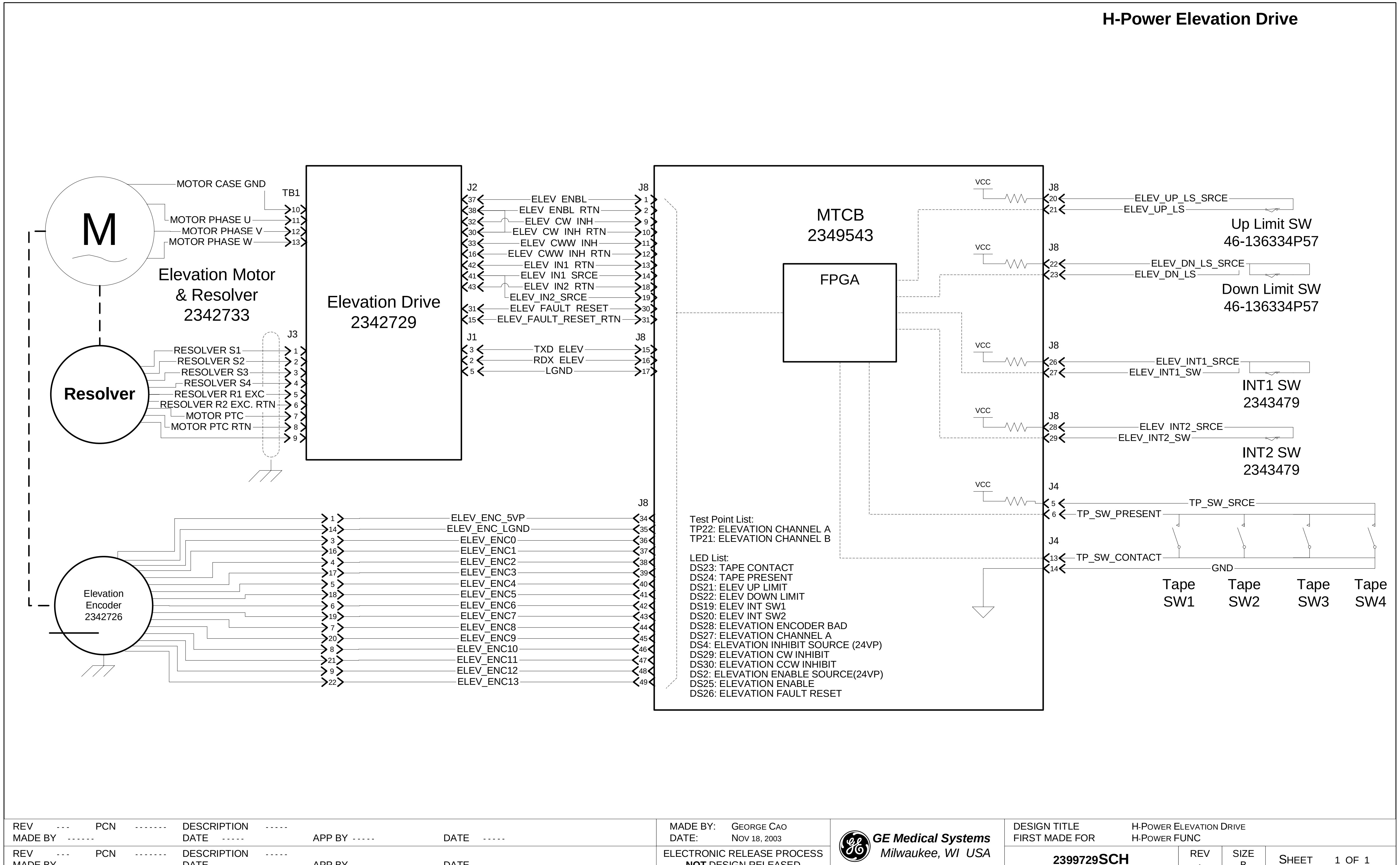
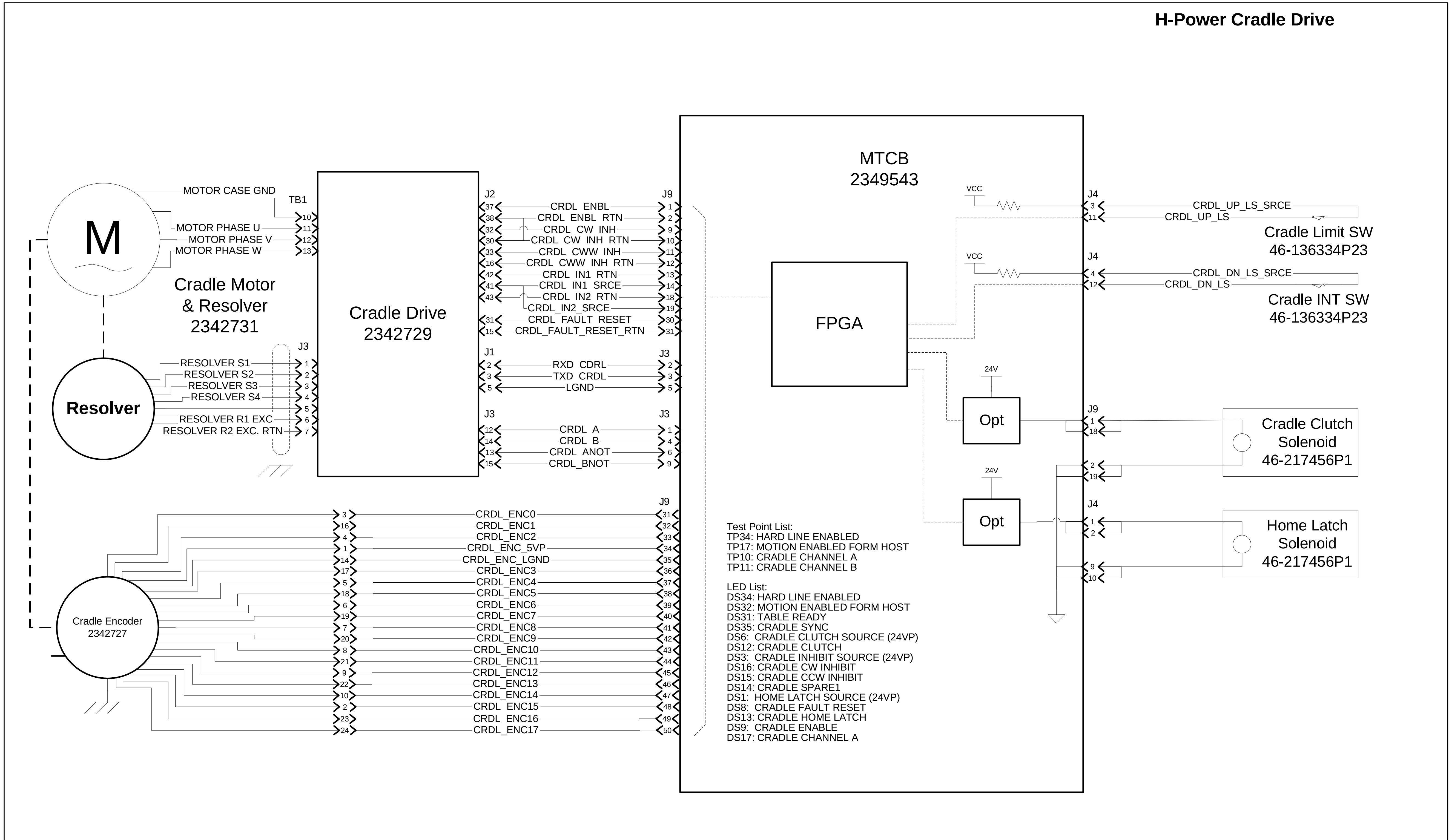


Figure 4-4 2399729 Sheet 1, Rev. 1

Section 5.0 — Table Cradle Drive - 2399730



REV --- PCN ------- DESCRIPTION ----- MADE BY ------ DATE ----- APP BY ----- DATE -----	MADE BY: GEORGE CAO DATE: NOV 18, 2003	GE Medical Systems Milwaukee, WI USA	DESIGN TITLE H-POWER CRADLE DRIVE FIRST MADE FOR H-POWER FUNC		
REV --- PCN ------- DESCRIPTION ----- MADE BY ------ DATE ----- APP BY ----- DATE -----	ELECTRONIC RELEASE PROCESS **NOT** DESIGN RELEASED		2399730SCH	REV 1	SIZE B / SHEET 1 OF 1

Figure 4-5 2399730 Sheet 1, Rev. 1

Chapter 5
Axial Control

Section 1.0 — Axial Servo Control / Dynamic Brake - 2399731

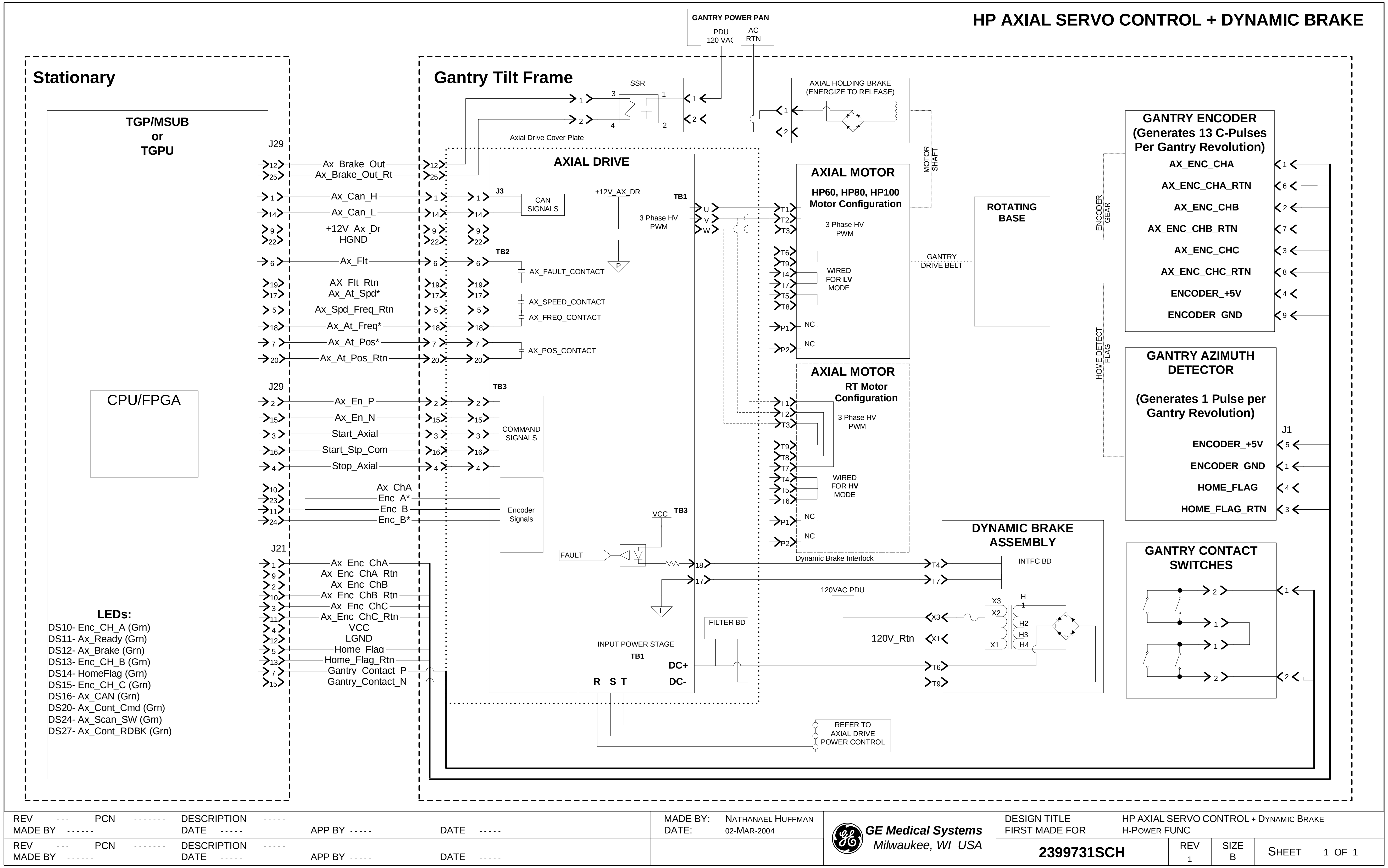
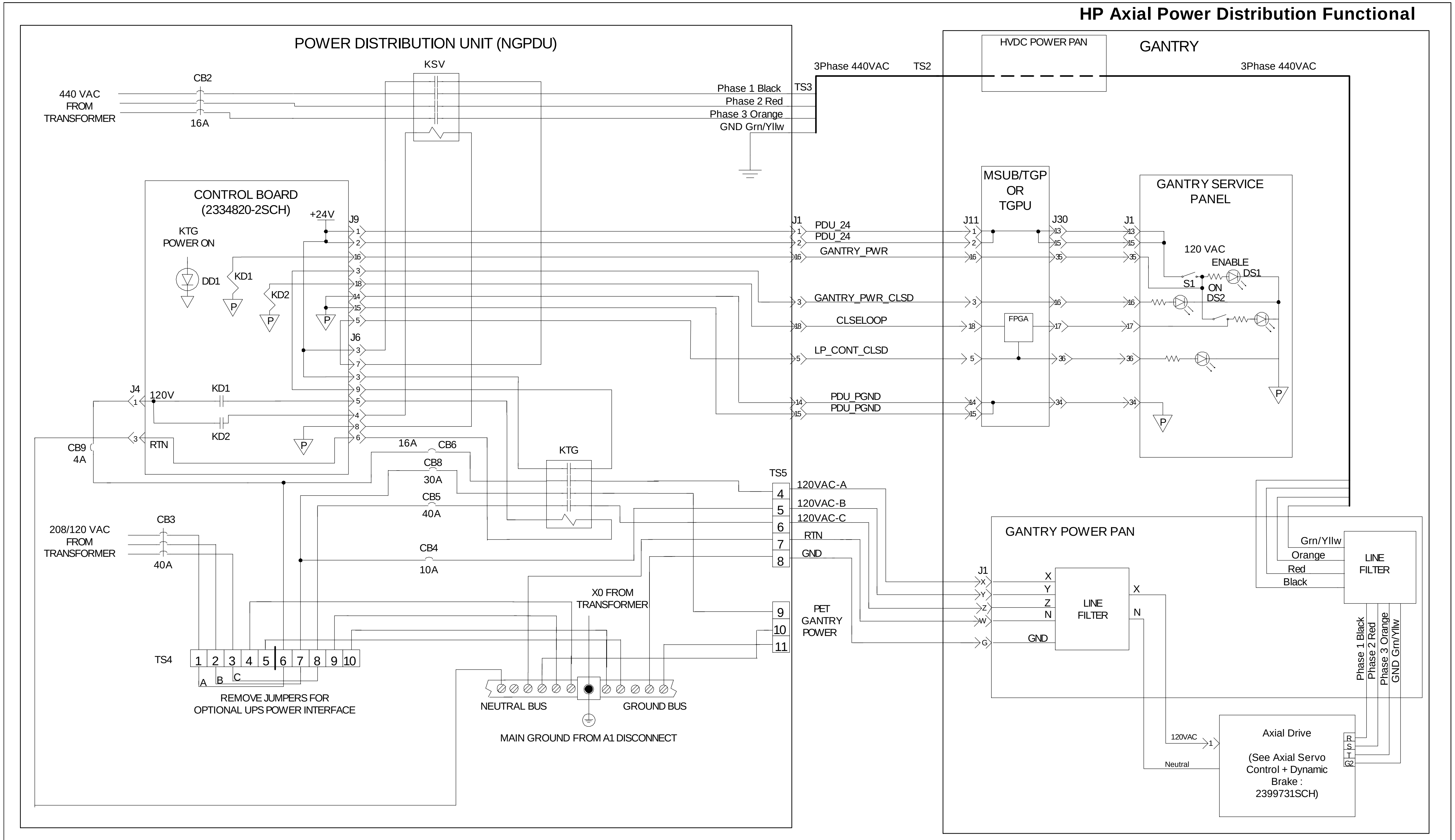


Figure 5-1 2399731 Sheet 1 Rev. 1

Section 2.0 — Axial Power Distribution - 2399732



REV ------ PCN ------- DESCRIPTION ---- -- MADE BY ------ DATE ------- APP BY DATE -----	MADE BY: ANDY PANOS DATE: 03/16/04	GE Medical Systems Milwaukee, WI USA	DESIGN TITLE HP AXIAL POWER DISTRIBUTION FUNCTIONAL FIRST MADE FOR H-POWER FUNC
REV 2 PCN ------- DESCRIPTION ADDED MORE DETAILS TO PDU AND MSUB PORTIONS. SEE PQR 13034162 MADE BY IGOR STEVIC DATE 03/01/2005 APP BY ----- DATE -----			2399732SCH REV 2 SIZE B SHEET 1 OF 1

Figure 5-2 2399732 Sheet 1 Rev. 2

THIS PAGE INTENTIONALLY BLANK.

Chapter 6
Operator I/O

Section 1.0 — Gantry Operator Interface and Displays - 2399733

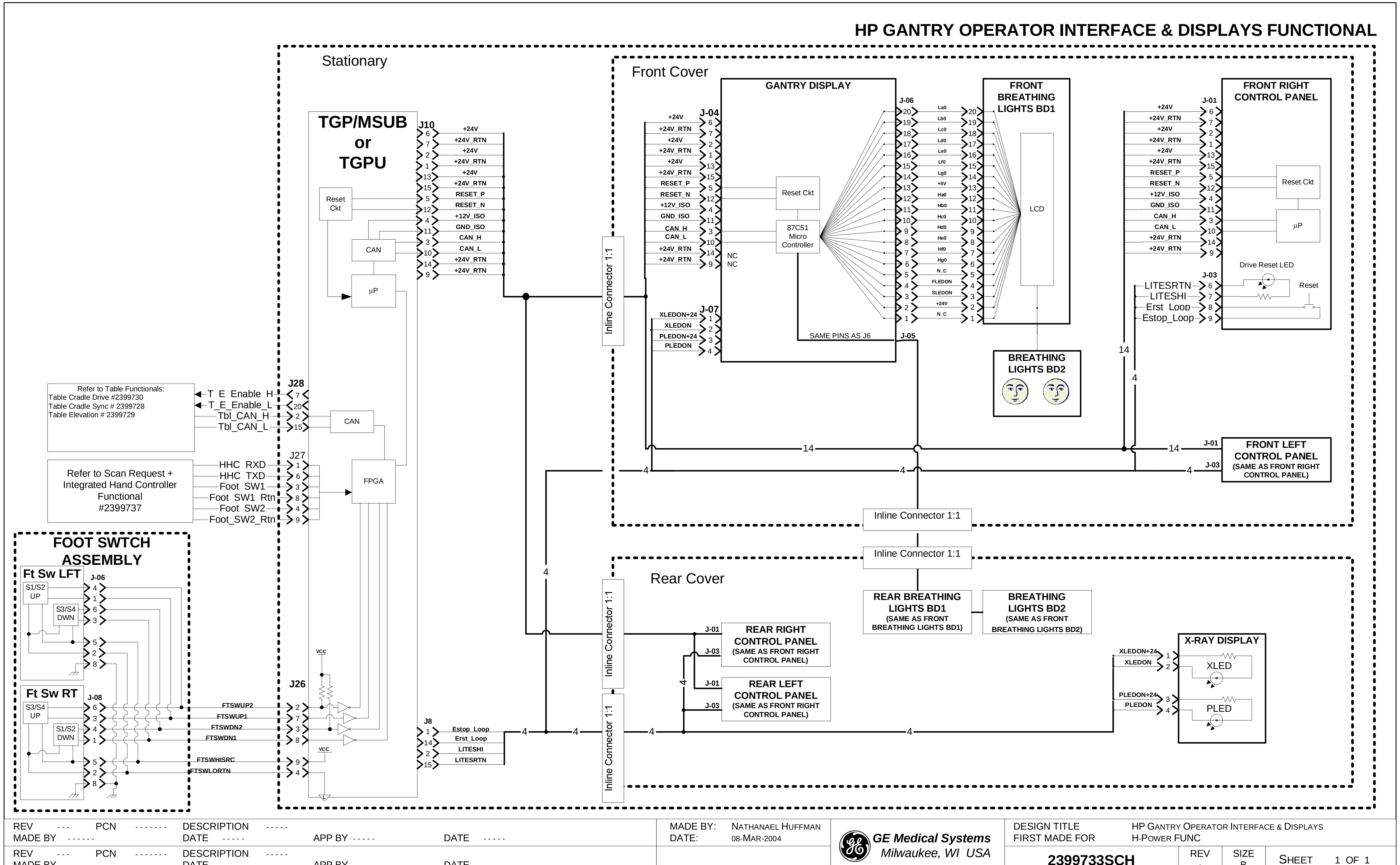
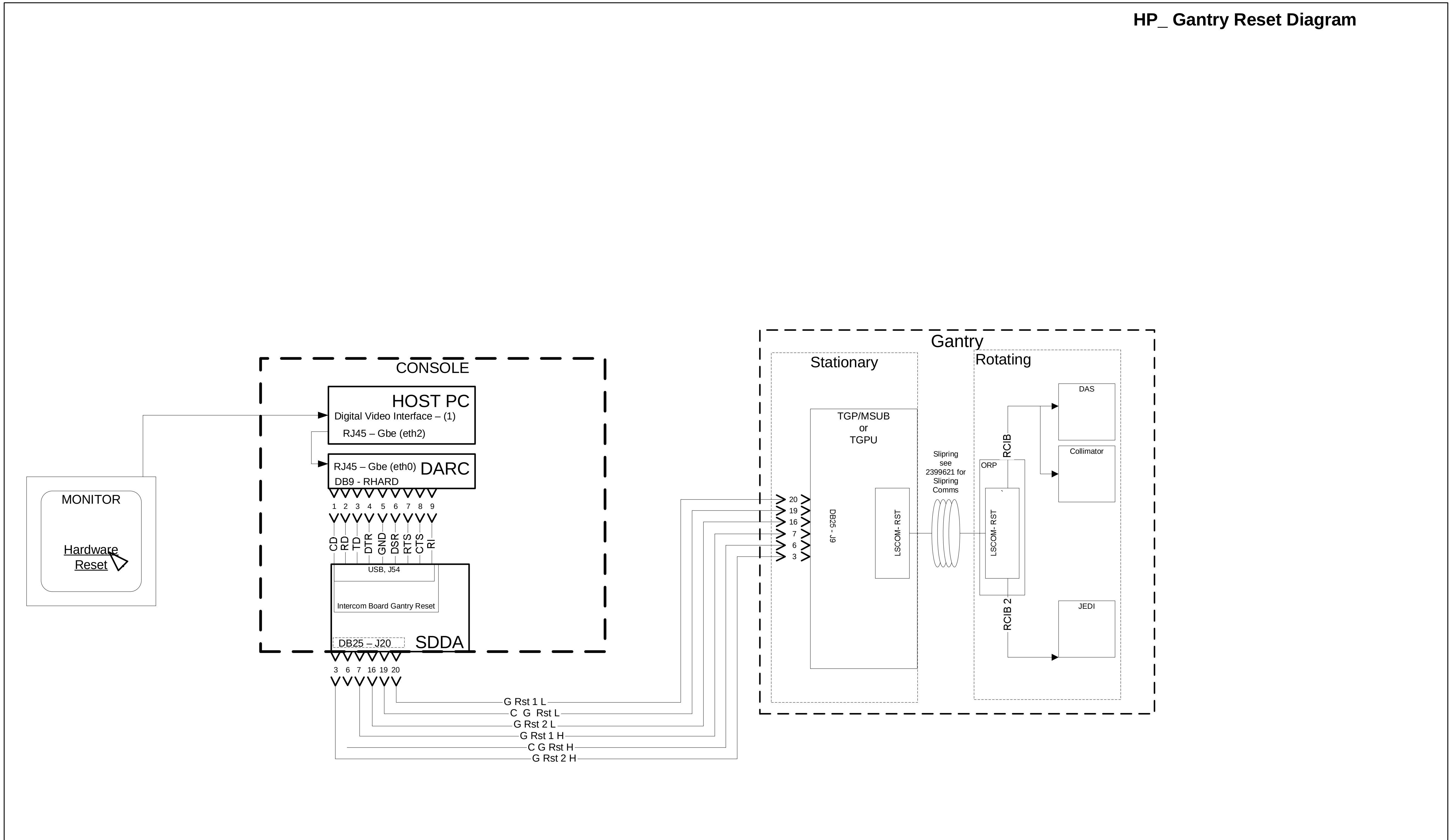


Figure 6-1 2399733 Sheet 1, Rev. 1

Section 2.0 — Gantry Reset from Console - 2399734



REV --- PCN ------- MADE BY ------	DESCRIPTION ----- DATE ----- APP BY ----- DATE -----	MADE BY: Nathanael Huffman DATE: 25-Feb-2004	GE Medical Systems Milwaukee, WI USA	DESIGN TITLE Console Gantry Reset FIRST MADE FOR H-Power FUNC	
REV --- PCN ------- MADE BY ------	DESCRIPTION ----- DATE ----- APP BY ----- DATE -----			2399734SCH	REV 1 SIZE B SHEET 1 OF 1

Figure 6-2 2399734 Sheet 1, Rev. 1

Section 3.0 — Cardiac Options - 2399735

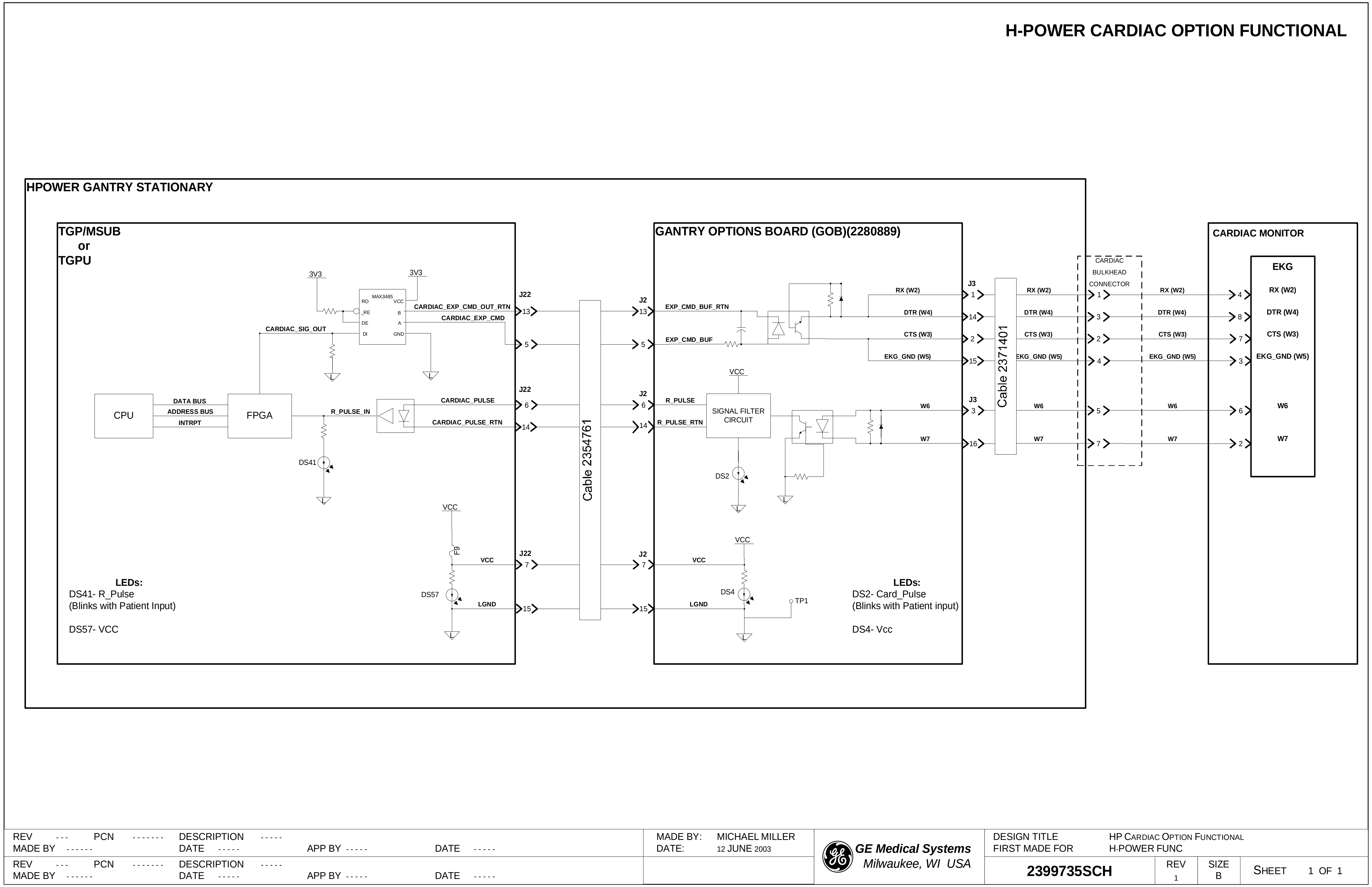


Figure 6-3 2399735 Sheet 1, Rev. 1

Section 4.0 — Pulmonary Options - 2399736

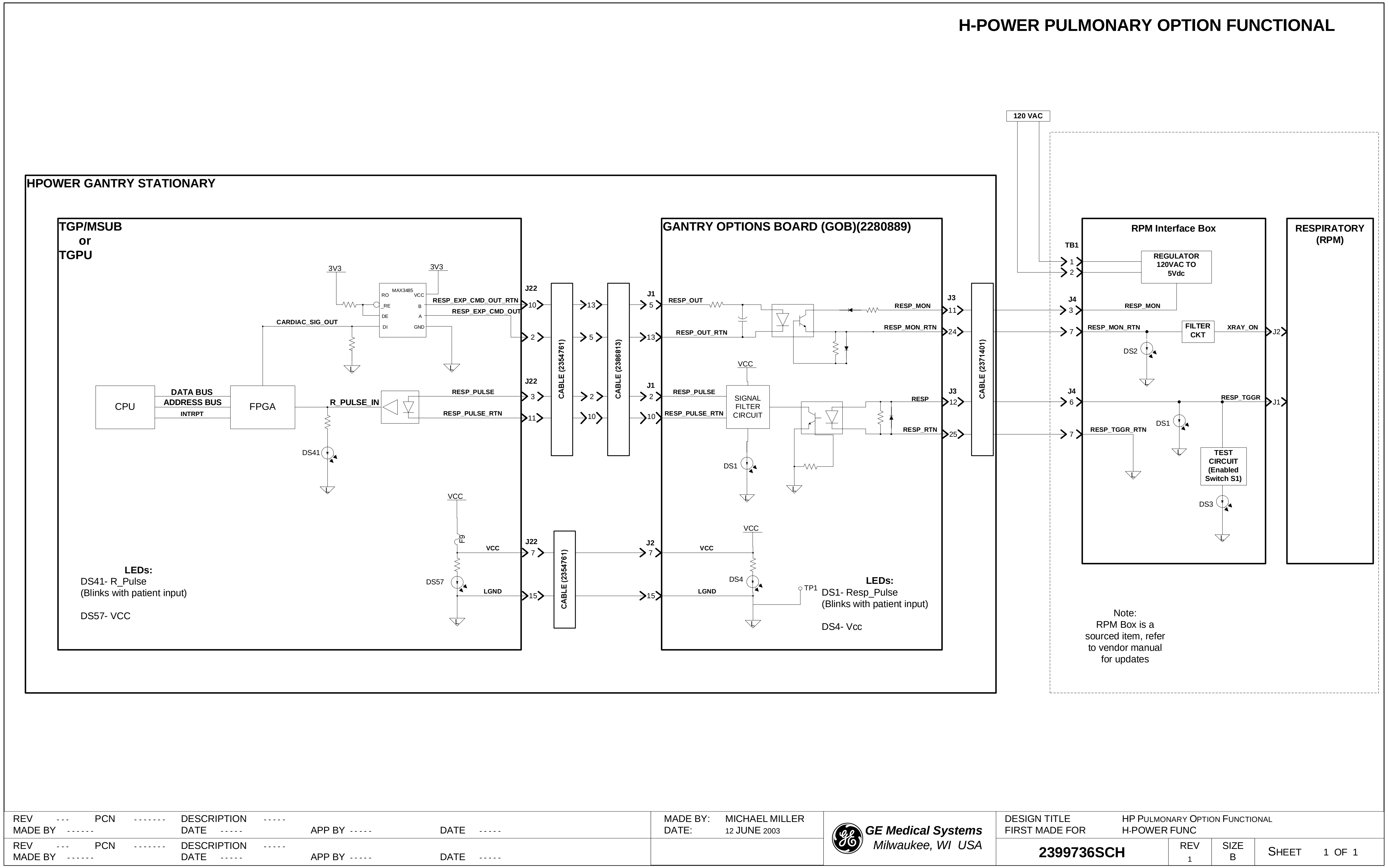


Figure 6-4 2399736 Sheet 1, Rev. 1

Section 5.0 — Scan Footswitch and IHHC Controller - 2399737

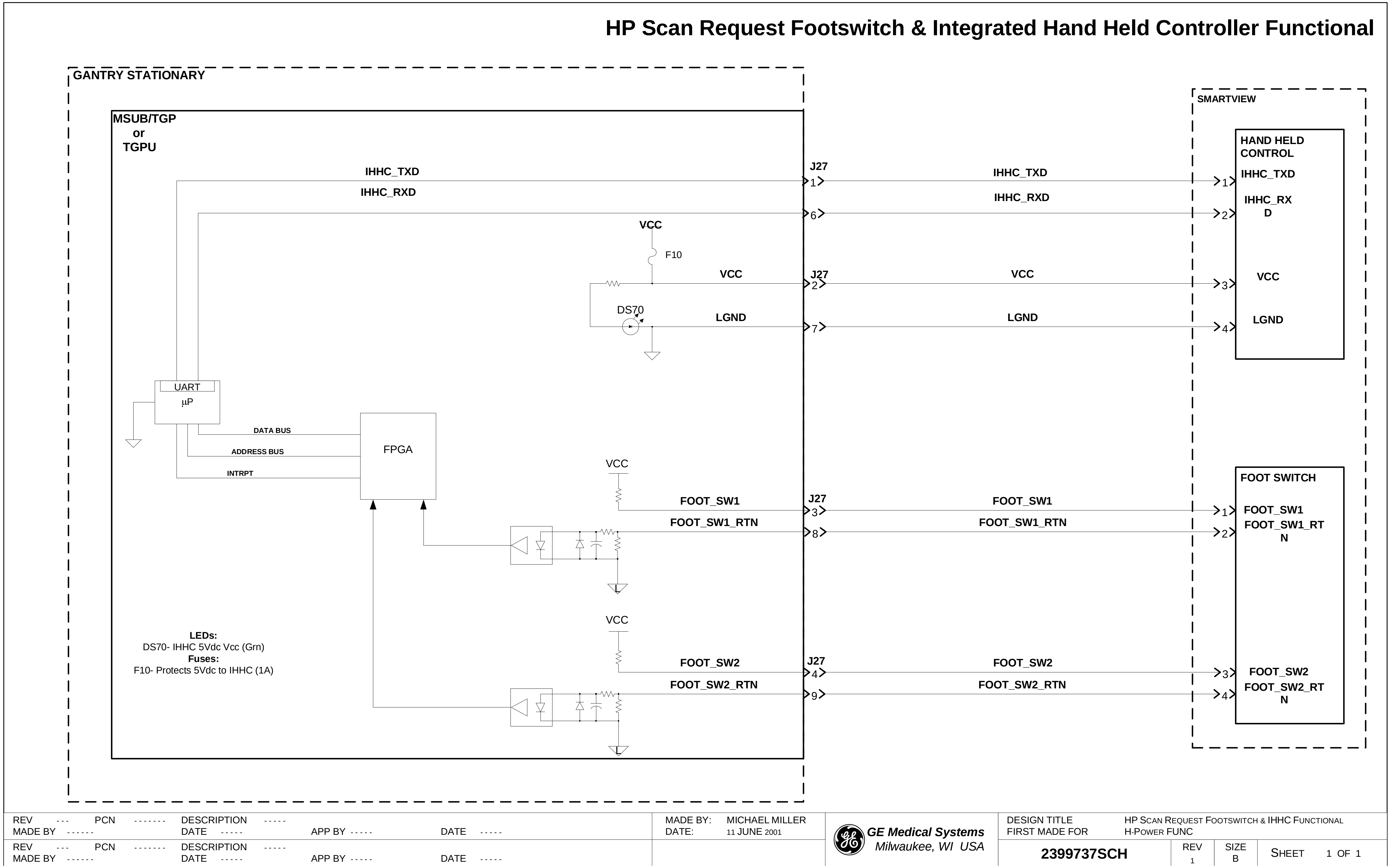
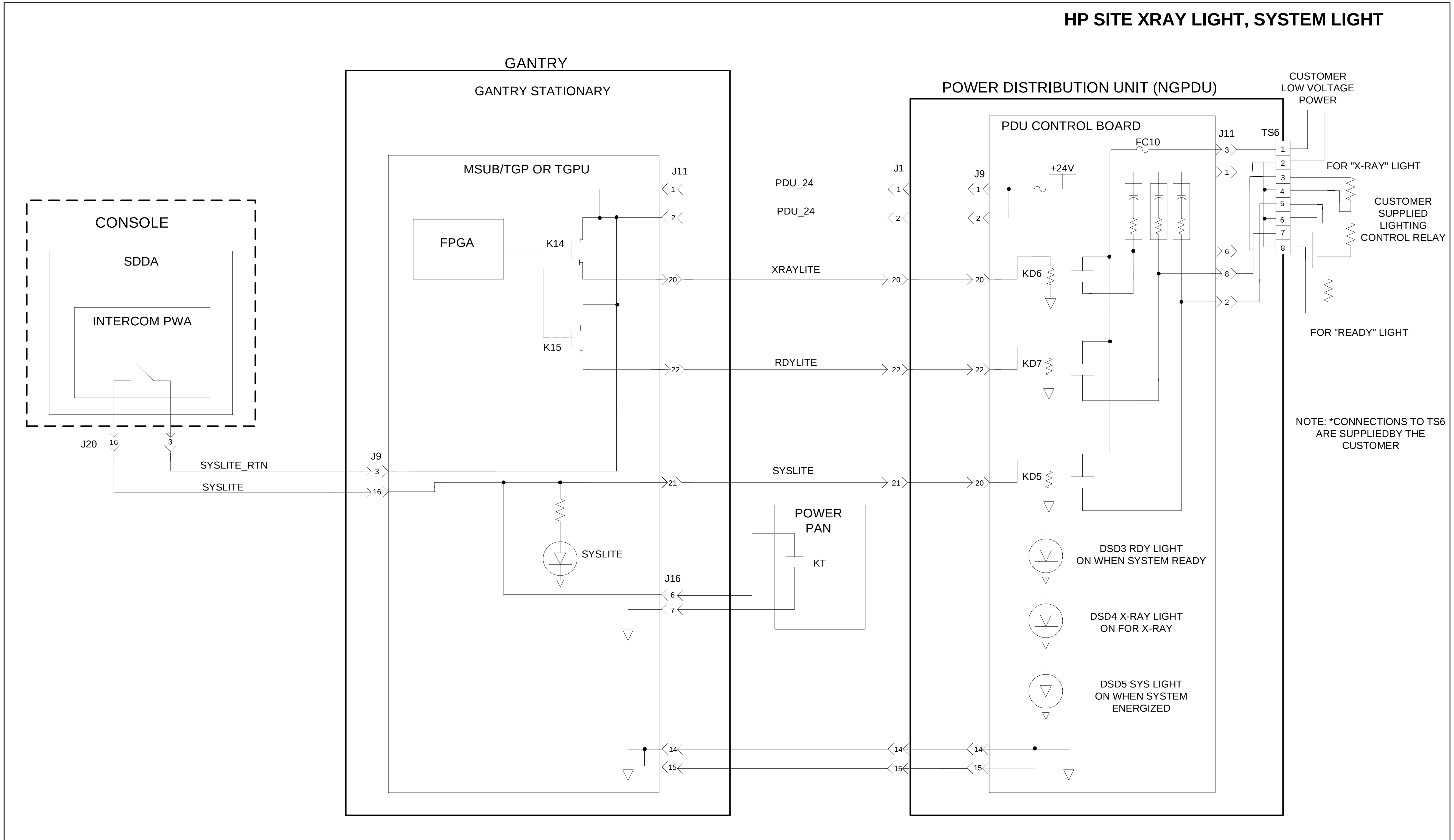


Figure 6-5 2399737 Sheet 1, Rev. 1

Section 6.0 — Site X-Ray Light, System Light - 2399738



REV --- PCN ------- DESCRIPTION ----- MADE BY ------ DATE ----- APP BY ----- DATE -----	MADE BY: ANDY PANOS DATE: 03/05/04	GE Medical Systems Milwaukee, WI USA	DESIGN TITLE HP SITE XRAY LIGHT, SYSTEM LIGHT FIRST MADE FOR H-POWER FUNC		
REV --- PCN ------- DESCRIPTION ----- MADE BY ------ DATE ----- APP BY ----- DATE -----			2399738SCH	REV 1	SIZE B	SHEET 1 OF 1

Figure 6-6 2399738 Sheet 1, Rev. 1

Section 7.0 — Autovoice / Intercom - 2399739

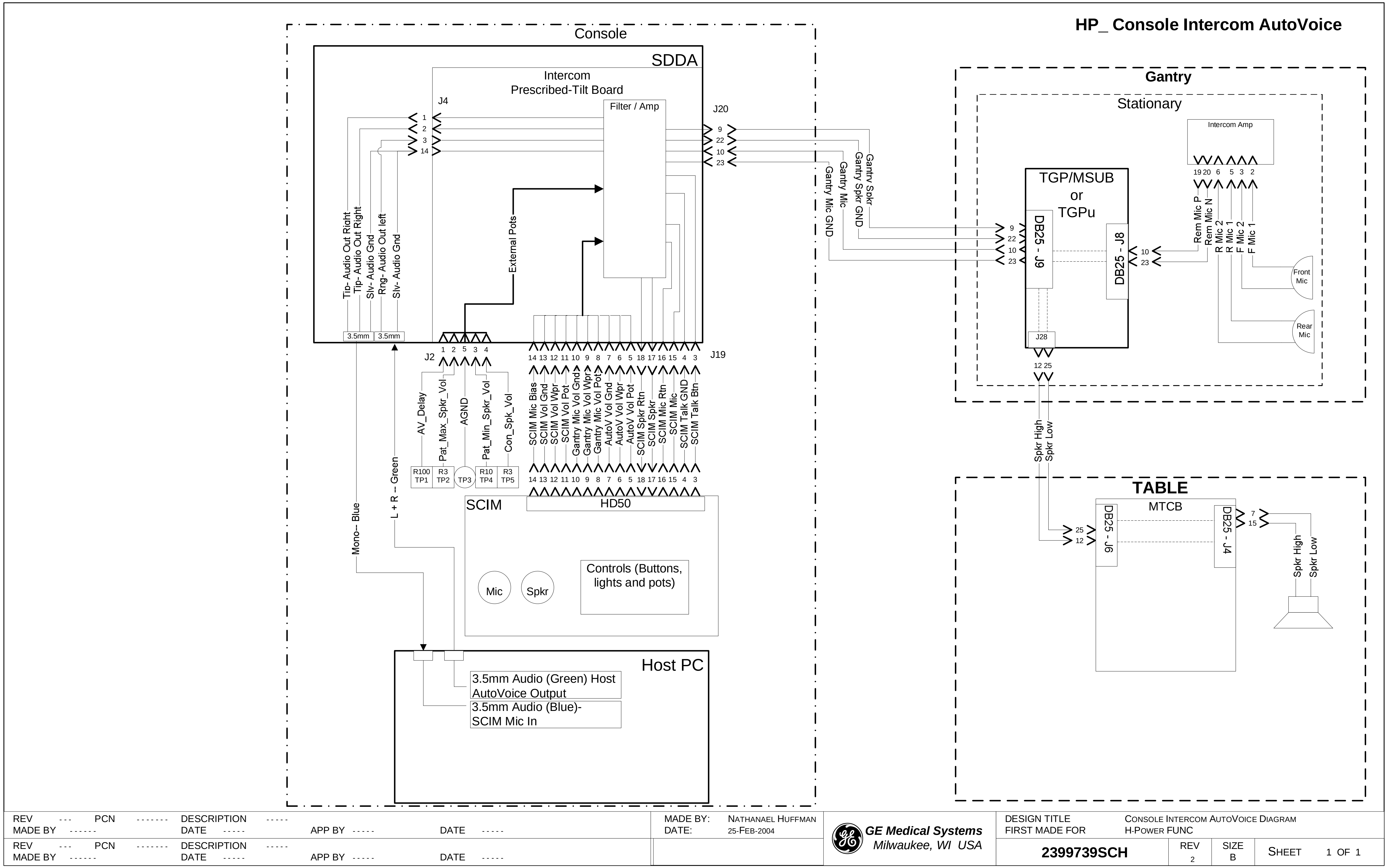


Figure 6-7 2399739 Sheet 1, Rev. 2

Chapter 7
System Monitoring

Section 1.0 — Gantry Temperature and Balance Sensors - 2399740

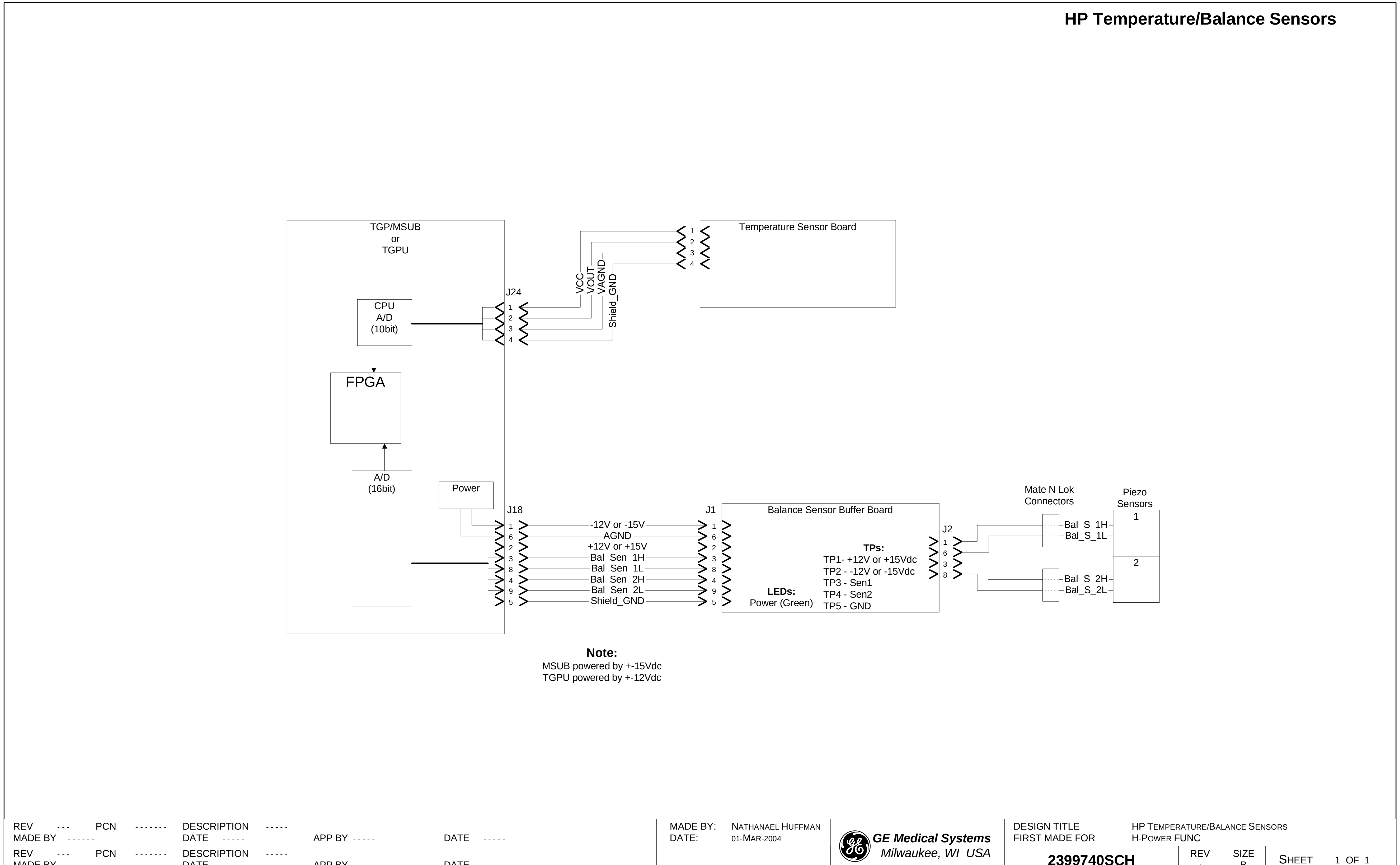
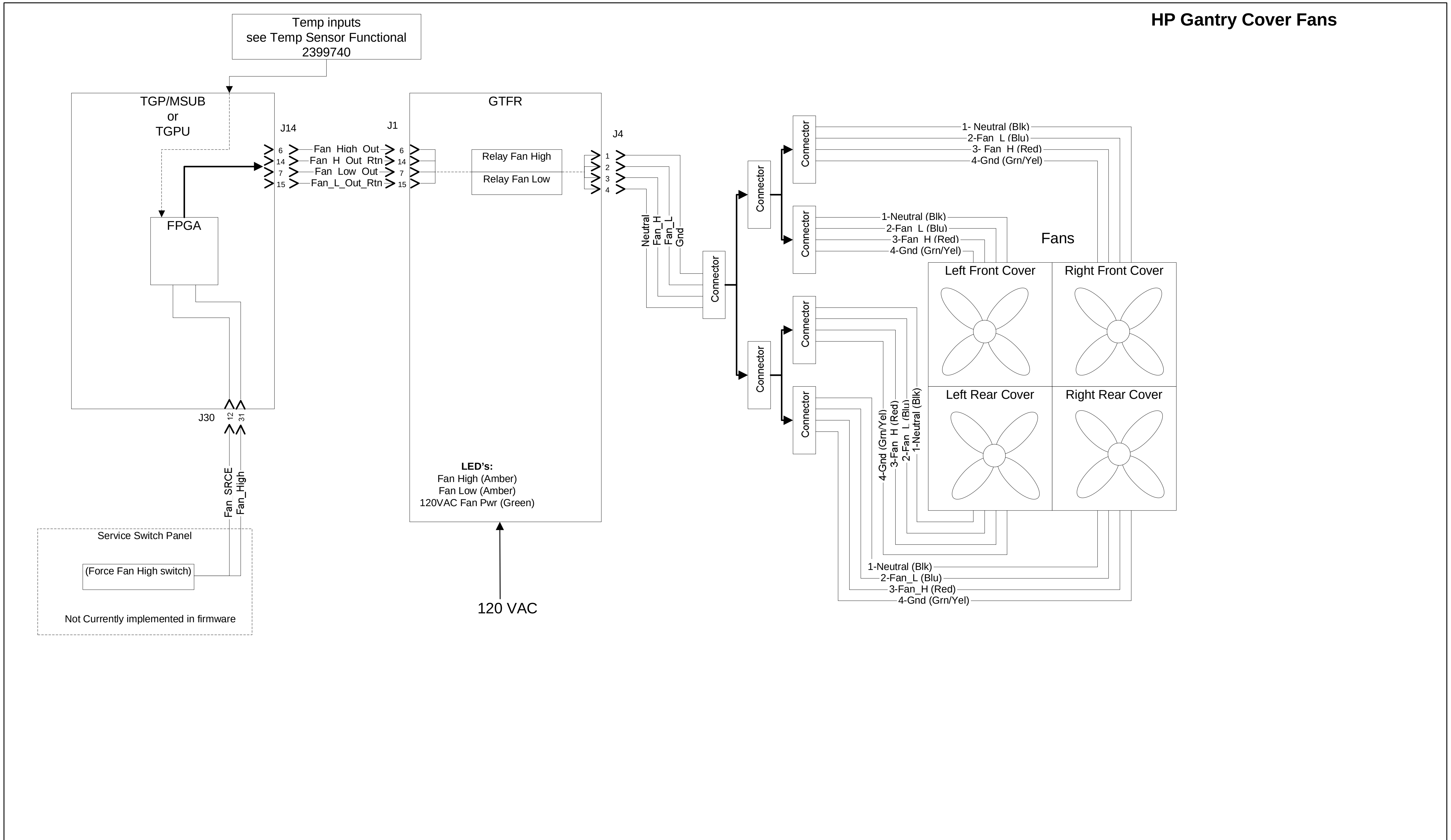


Figure 7-1 2399740 Sheet 1, Rev. 1

Section 2.0 — Gantry Cover Fans - 2399741



REV ---	PCN -------	DESCRIPTION -----			MADE BY: NATHANAEL HUFFMAN	GE Medical Systems Milwaukee, WI USA	DESIGN TITLE	HP GANTRY COVER FANS
MADE BY ------		DATE -----	APP BY -----	DATE -----	DATE: 23-FEB-2004		FIRST MADE FOR	H-POWER FUNC
REV ---	PCN -------	DESCRIPTION -----					2399741SCH	REV 1 · SIZE B · SHEET 1 OF 1
MADE BY ------		DATE -----	APP BY -----	DATE -----				

Figure 7-2 2399741 Sheet 1, Rev. 1

THIS PAGE INTENTIONALLY BLANK.

Chapter 8
System Power Control

Section 1.0 — System E-Stop and Power Interlocks - 2399742

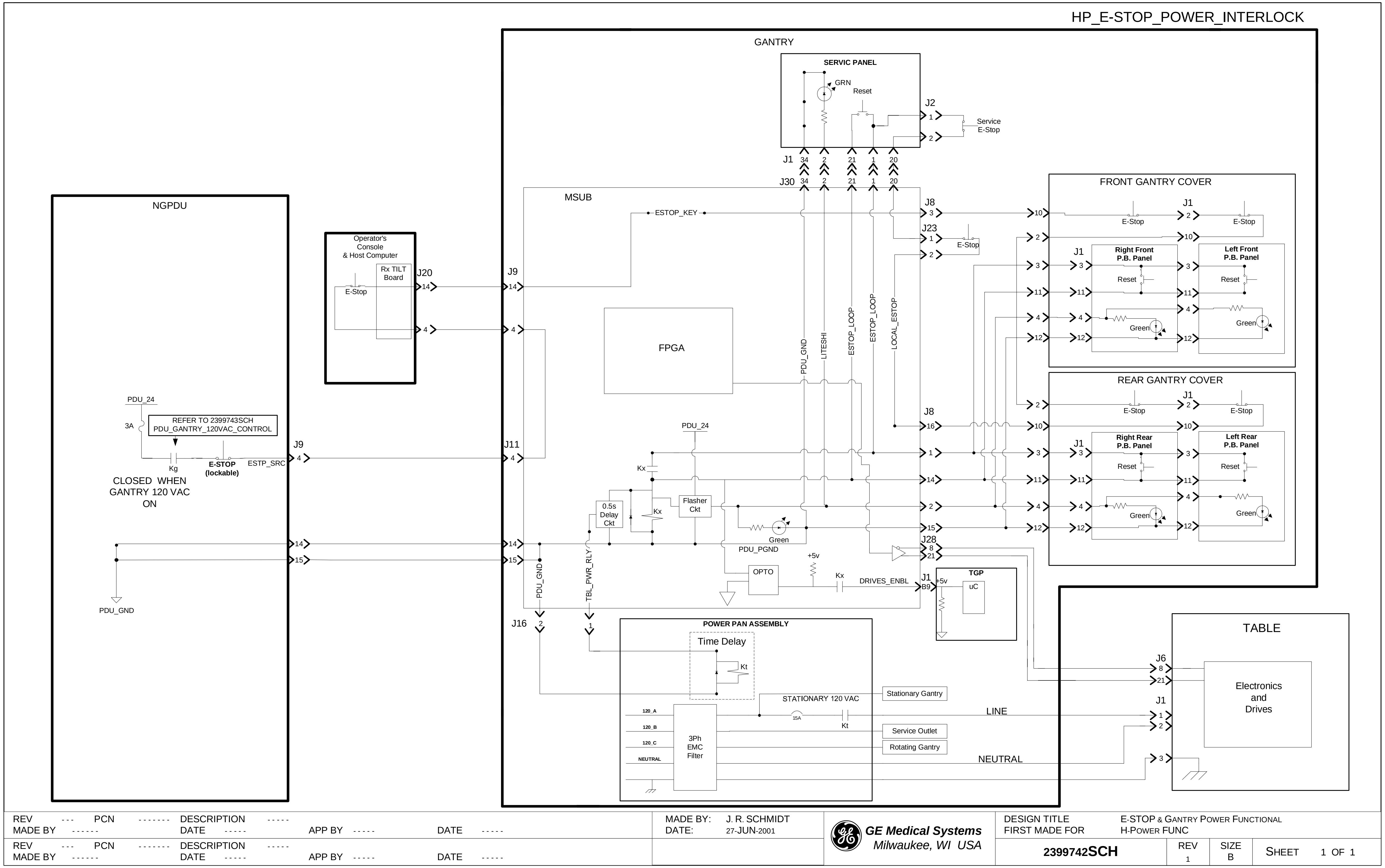
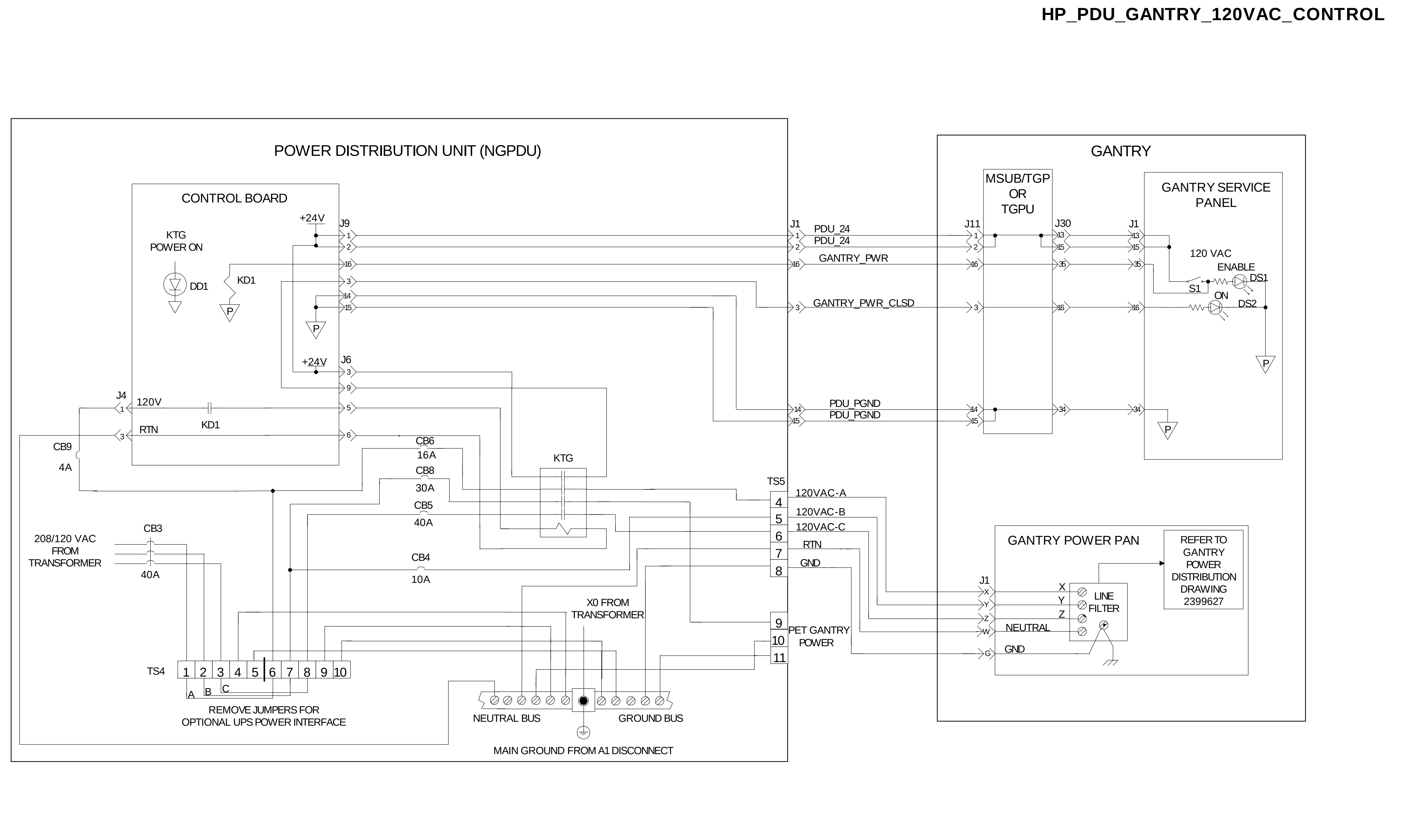


Figure 8-1 2399742 Sheet 1, Rev. 1

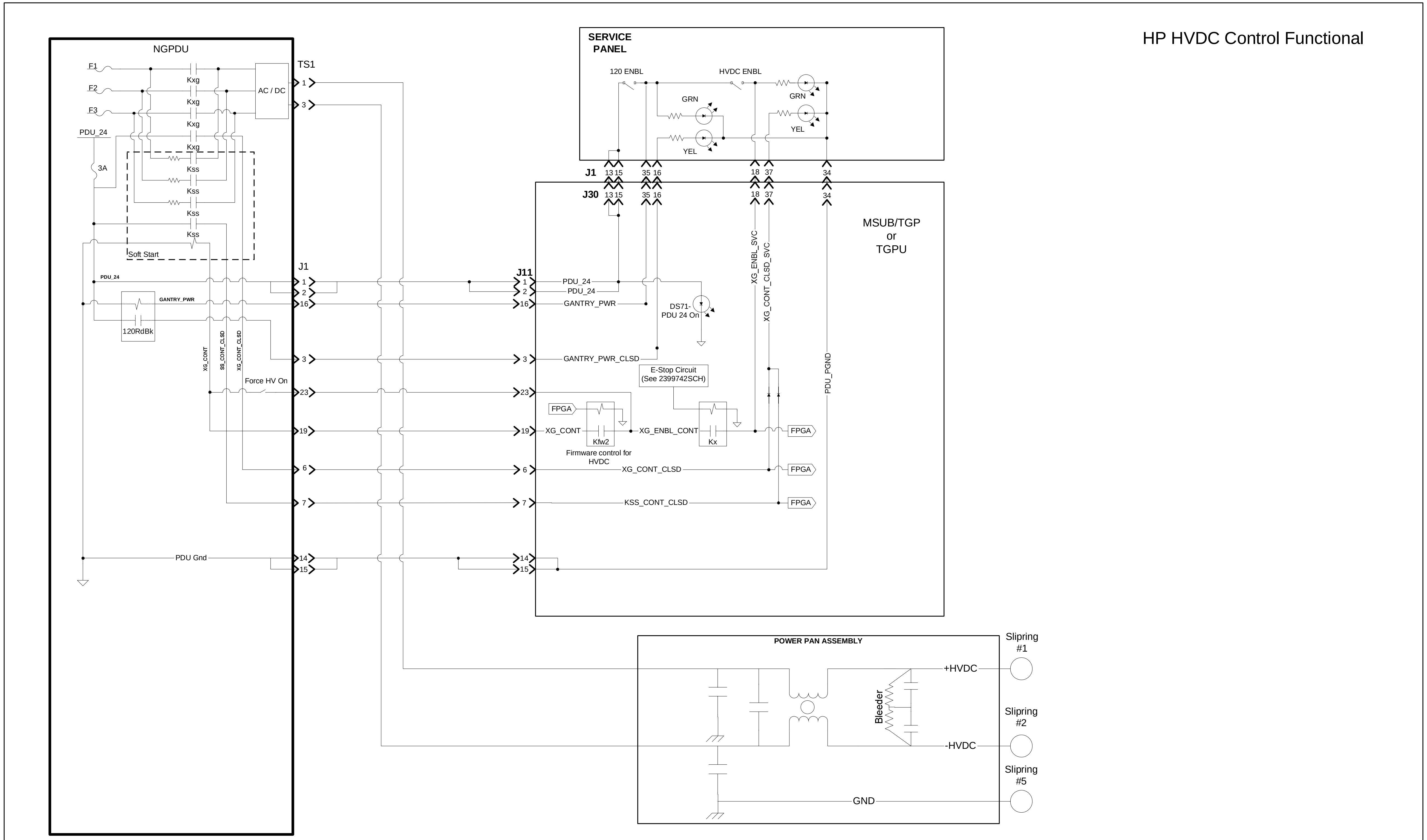
Section 2.0 — PDU - Gantry 120 VAC Control - 2399743



REV - - - PCN - - - - - - - DESCRIPTION - - - - - MADE BY - - - - - - DATE - - - - - APP BY - - - - - DATE - - - - -	MADE BY: ANDY PANOS DATE: 03/16/04	GE Medical Systems Milwaukee, WI USA	DESIGN TITLE HP_PDU_GANTRY_120VAC_CONTROL FIRST MADE FOR H-POWER FUNC		
REV 3 PCN - - - - - - - DESCRIPTION ADDED LABELS FOR TS2 AND TS5. ADDED SCHEMATIC NUMBER FOR GANTRY PWR DIST MADE BY IGOR STEVIC DATE 02/20/2005 APP BY - - - - - DATE - - - - -			2399743SCH	REV 3	SIZE B	SHEET 1 OF 1

Figure 8-2 2399743 Sheet 1, Rev. 3

Section 3.0 — HV_DC_Control - 2399744



REV --- MADE BY ------	PCN -------	DESCRIPTION ----- DATE -----	APP BY -----	DATE -----	MADE BY: NATHANAEL HUFFMAN DATE: 01-APR-2004
REV --- MADE BY ------	PCN -------	DESCRIPTION ----- DATE -----	APP BY -----	DATE -----	

GE Medical Systems
Milwaukee, WI USA

DESIGN TITLE	HP HVDC CONTROL FUNCTIONAL			
FIRST MADE FOR	H-POWER FUNC			
2399744SCH	REV 1	SIZE B	SHEET	1 OF 1

Figure 8-3 2399744 Sheet 1, Rev. 1

Section 4.0 — HP MDAS/GDAS Gantry Power Distribution - 2399627

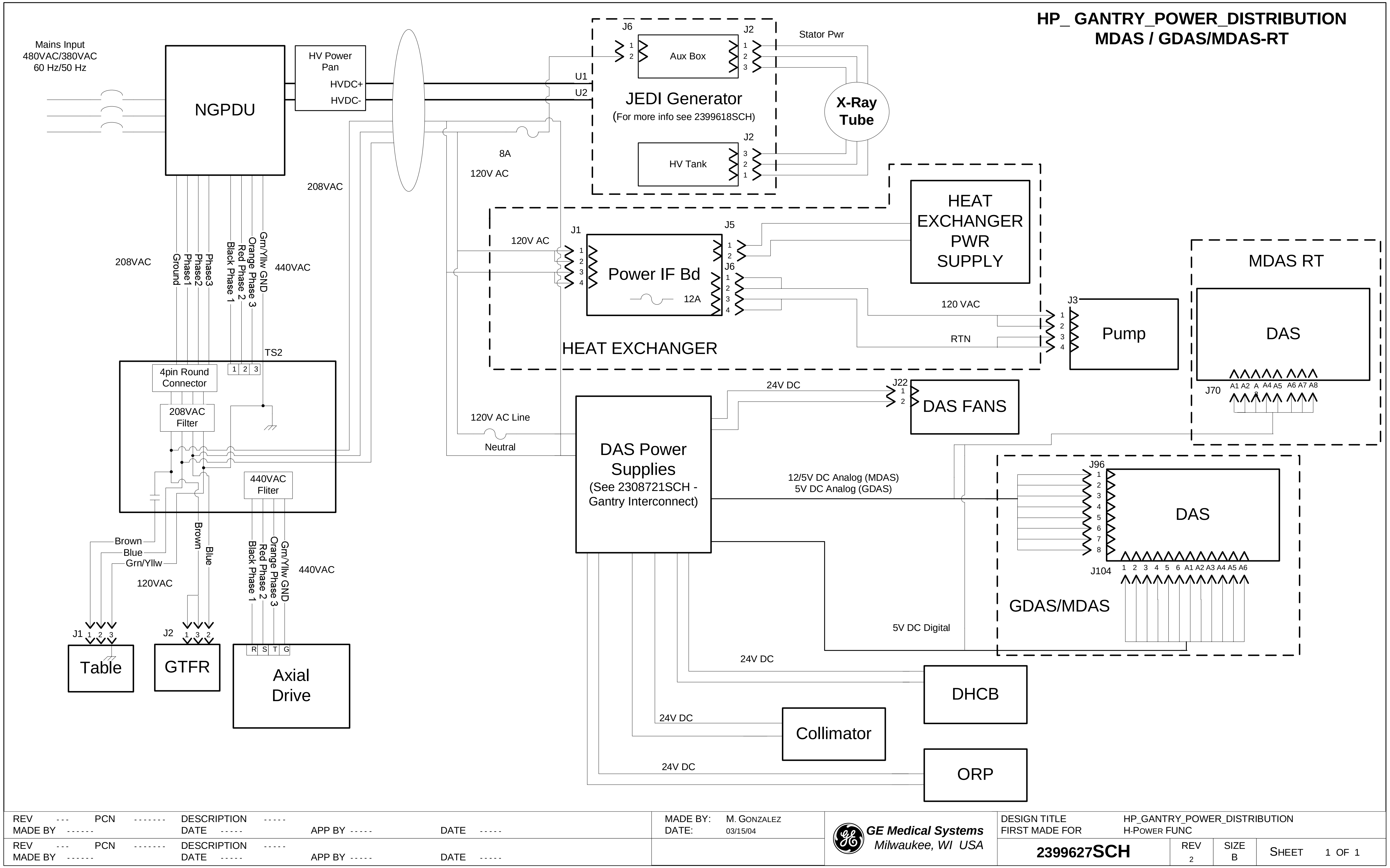


Figure 8-4 2399627 Sheet 1, Rev. 2

THIS PAGE INTENTIONALLY BLANK.

GE Medical Systems, a General Electric Company, going to market as GE Healthcare.
3000 N. Grandview Boulevard
Waukesha, Wisconsin 53188
USA

www.gehealthcare.com